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(54) **SMALL FORM-FACTOR DEVICES WITH INTEGRATED AND MODULAR HARVESTING RECEIVERS, AND SHELVING-MOUNTED WIRELESS-POWER TRANSMITTERS FOR USE THEREWITH**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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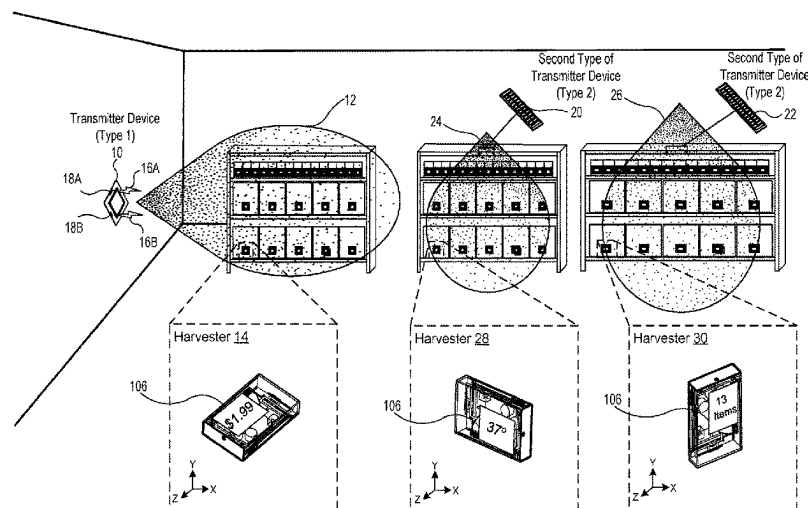
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(57) **ABSTRACT**

A wireless-power harvester integrated in a small device, comprising a stamped metal harvesting antenna. The stamped metal antenna is formed into a meandering shape. A first end of the meandering shape is a free end positioned within free space of a housing of a small device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small device. The PCB is configured to operate as a ground plane for the stamped metal antenna. An intermediate portion, disposed between the first end and the second end of the meandering shape, is coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small device or for powering the small device.

14 Claims, 16 Drawing Sheets



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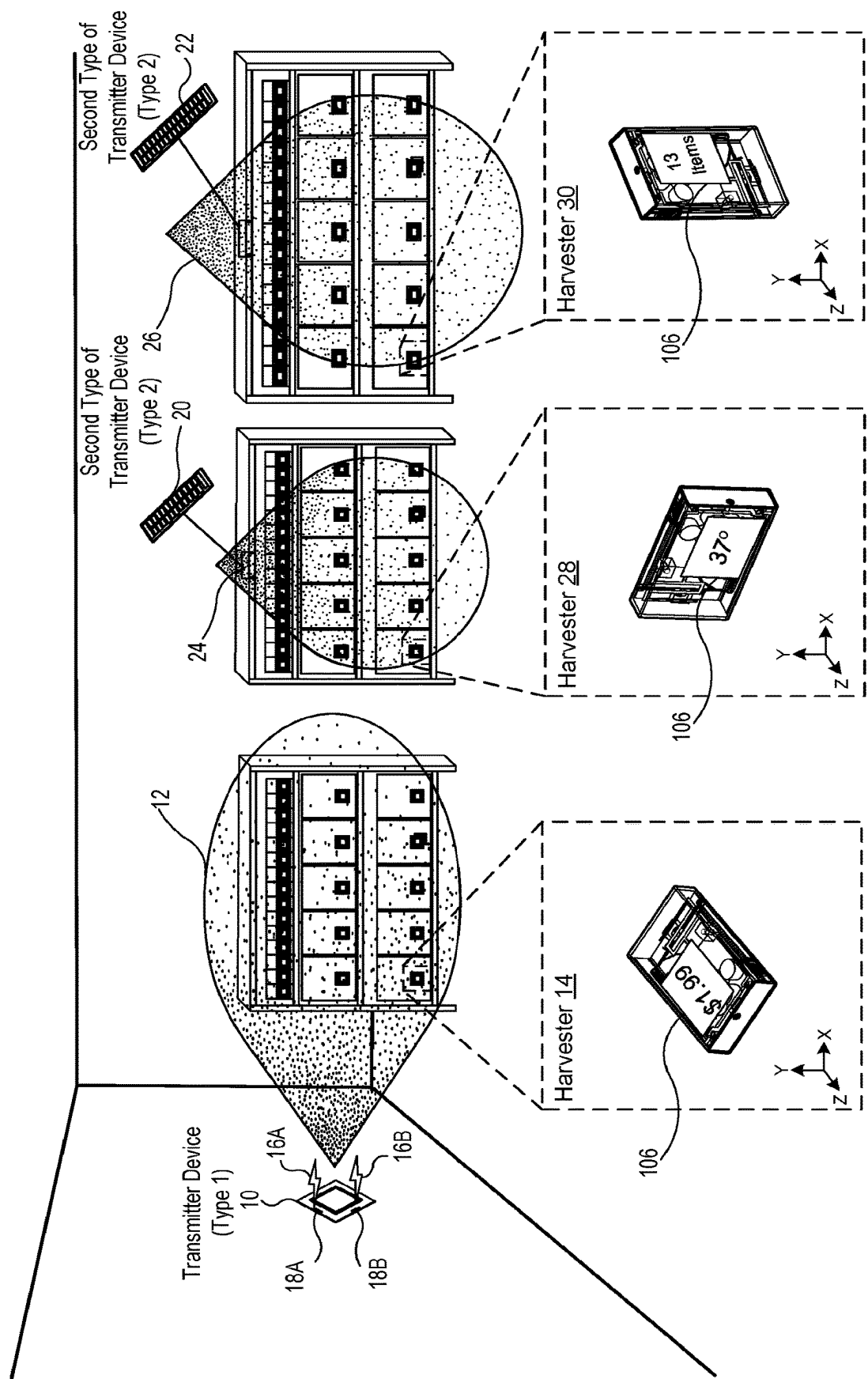


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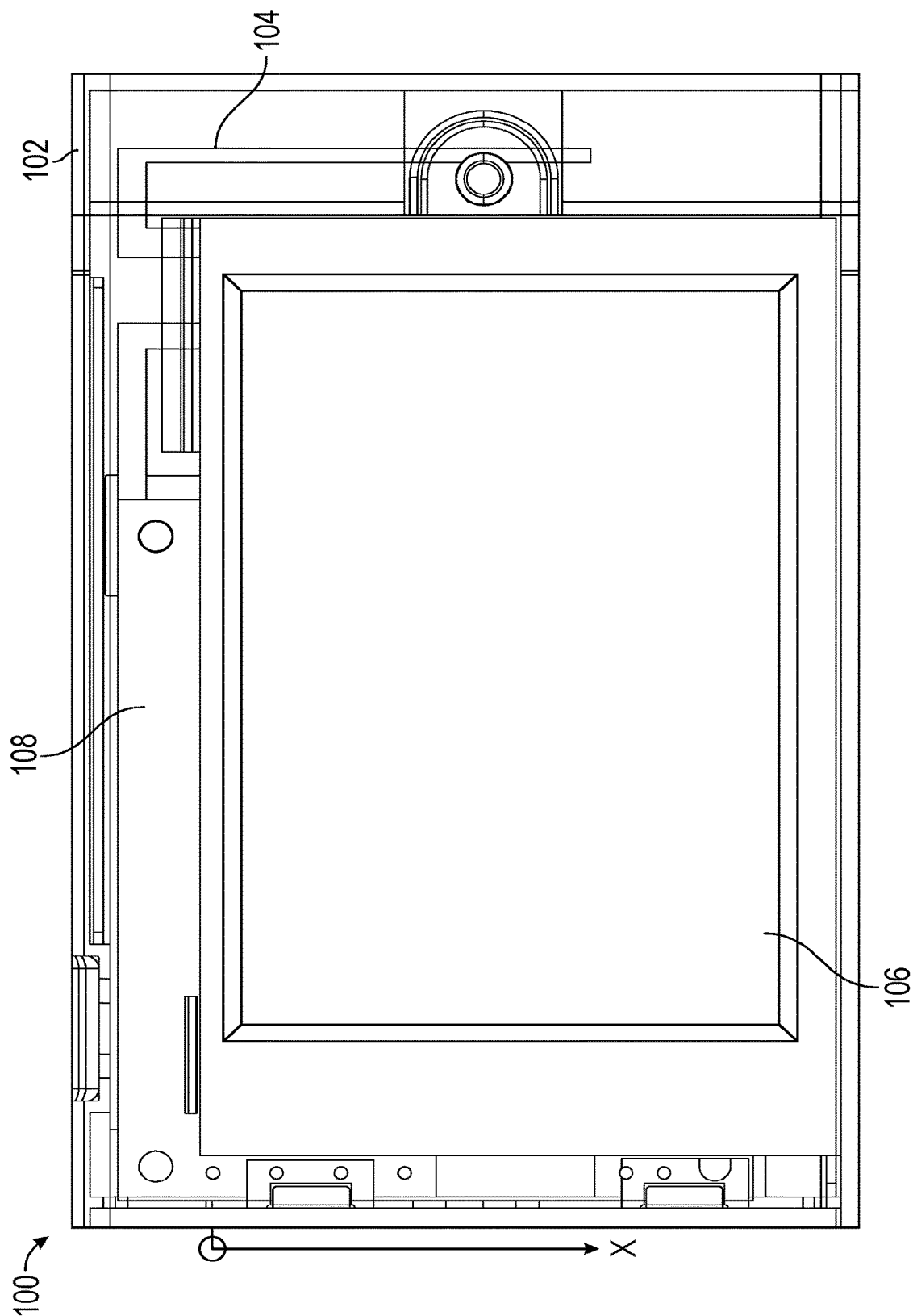


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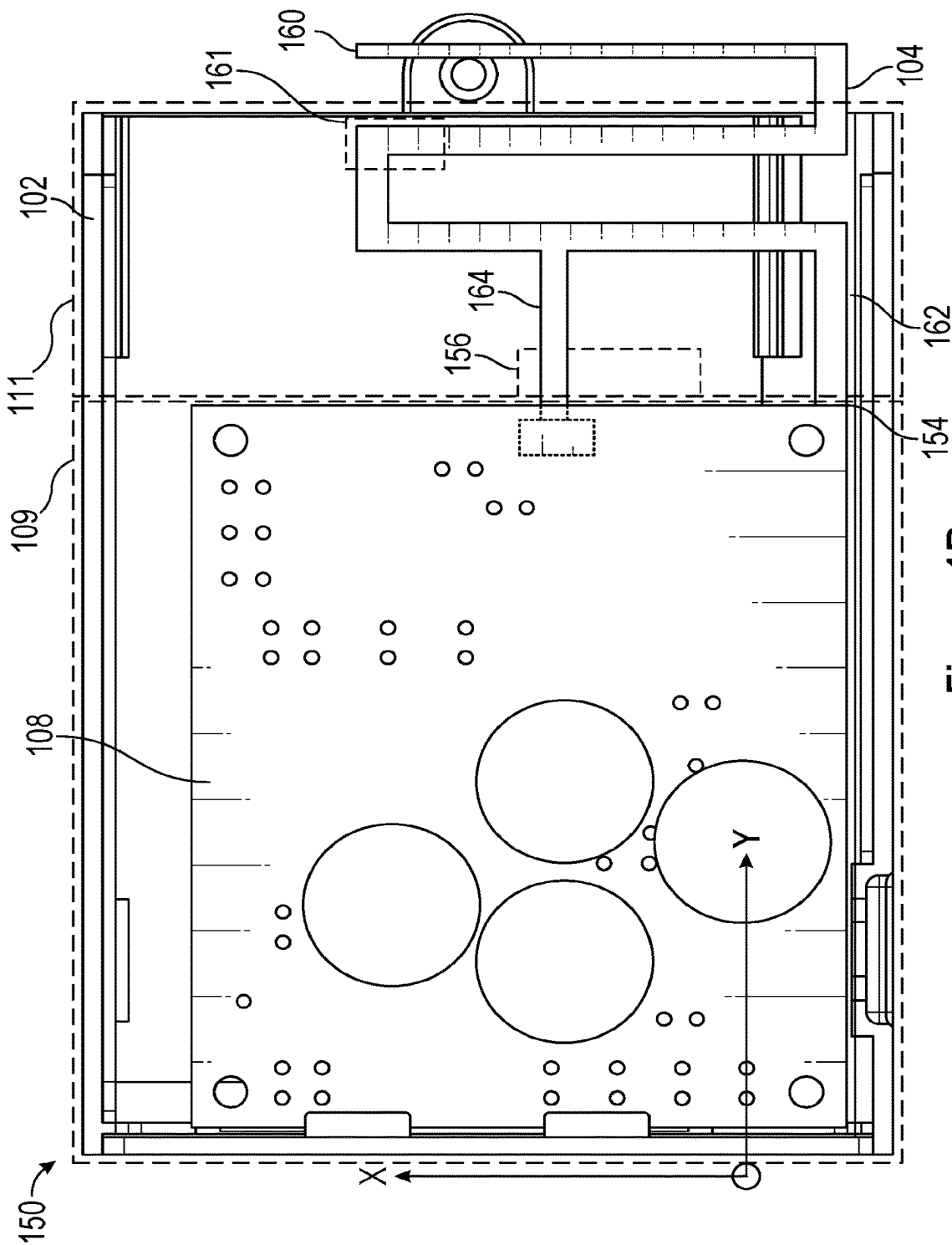


Figure 1B

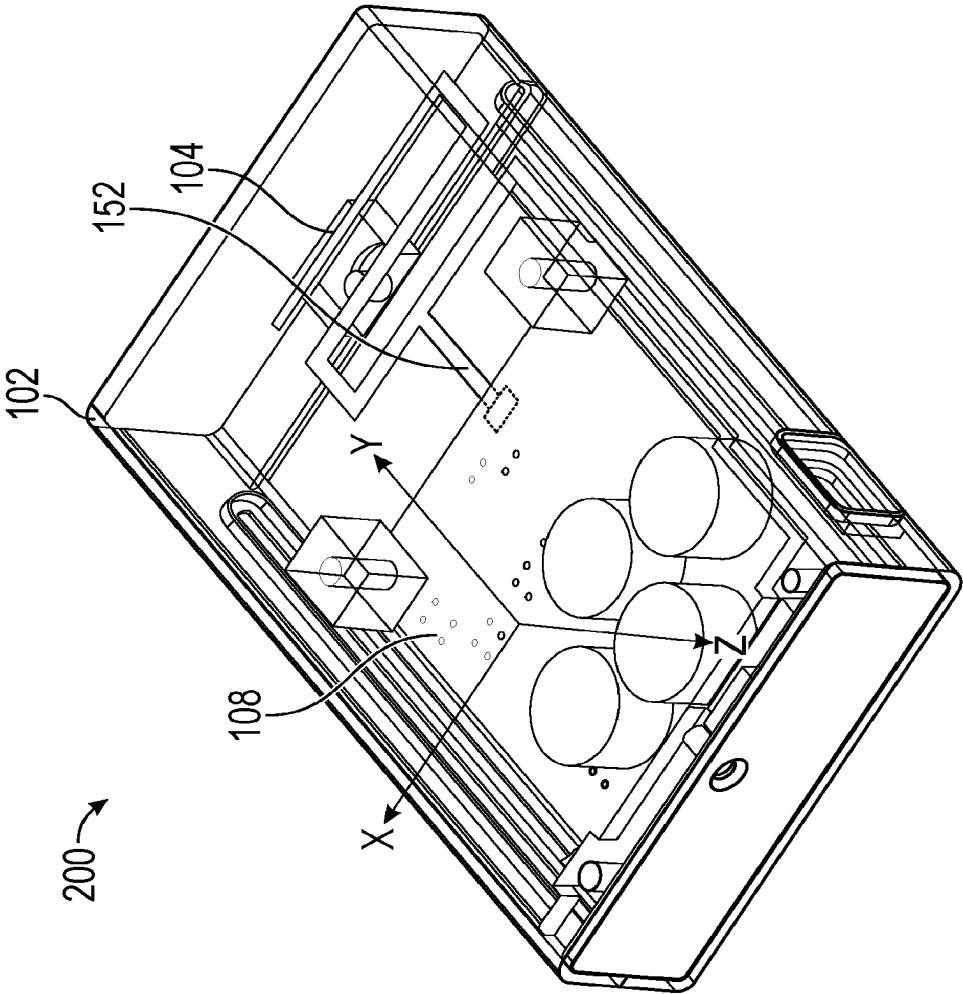


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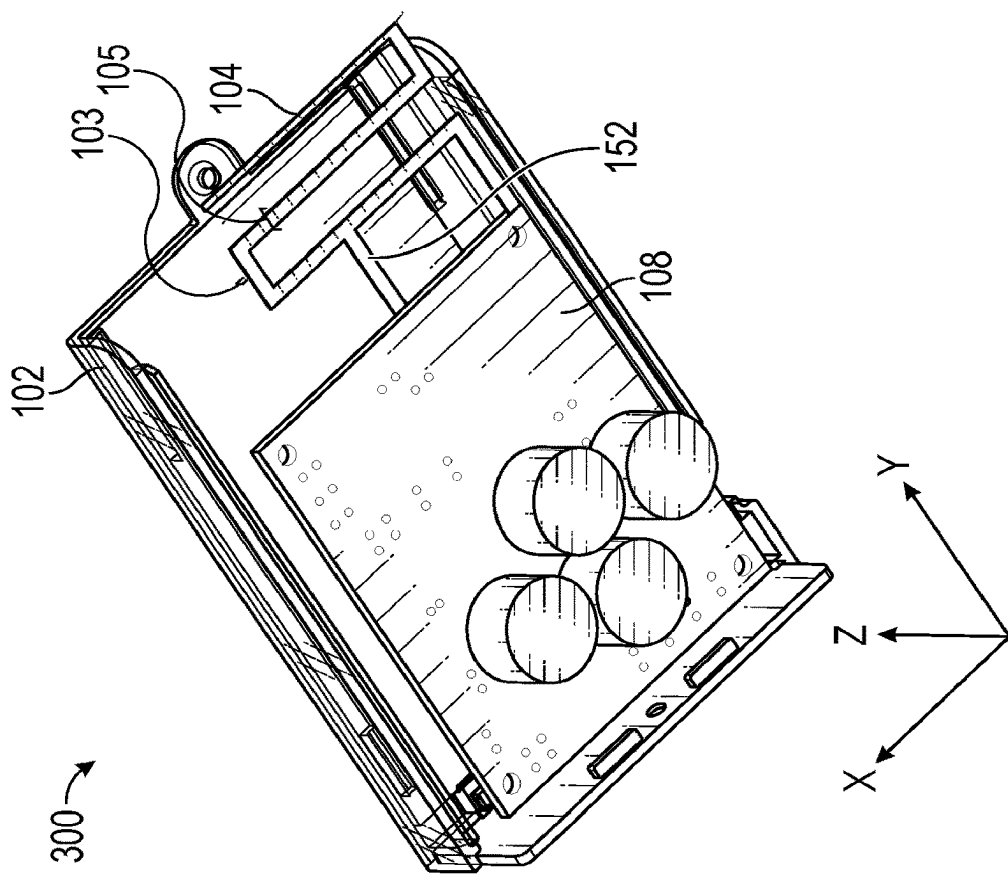


Figure 3A

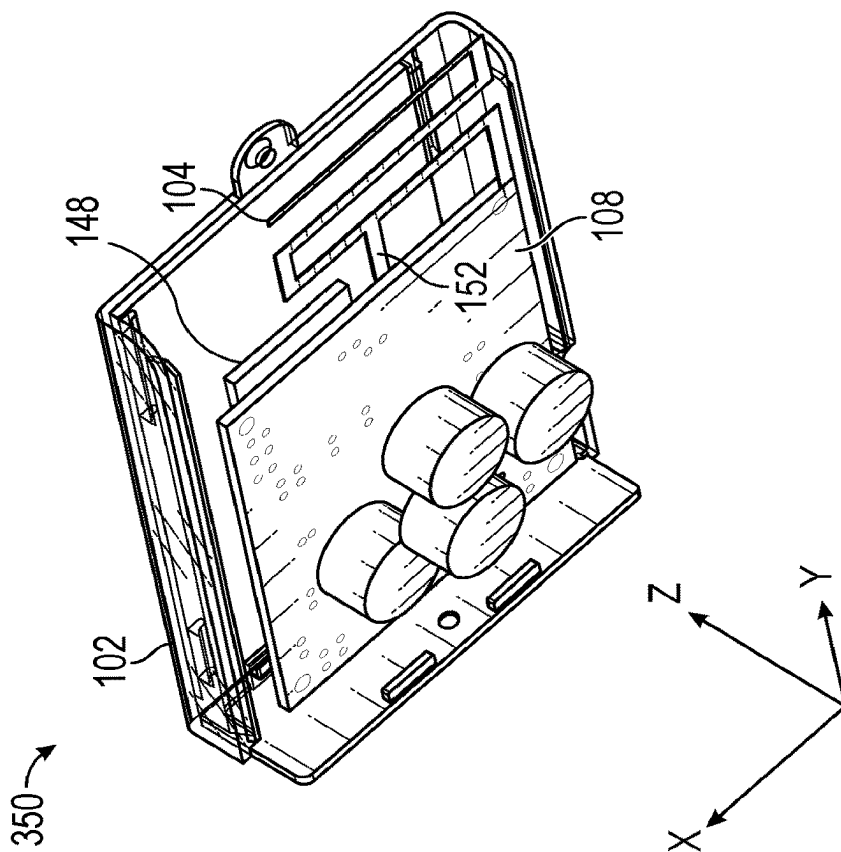


Figure 3B

400

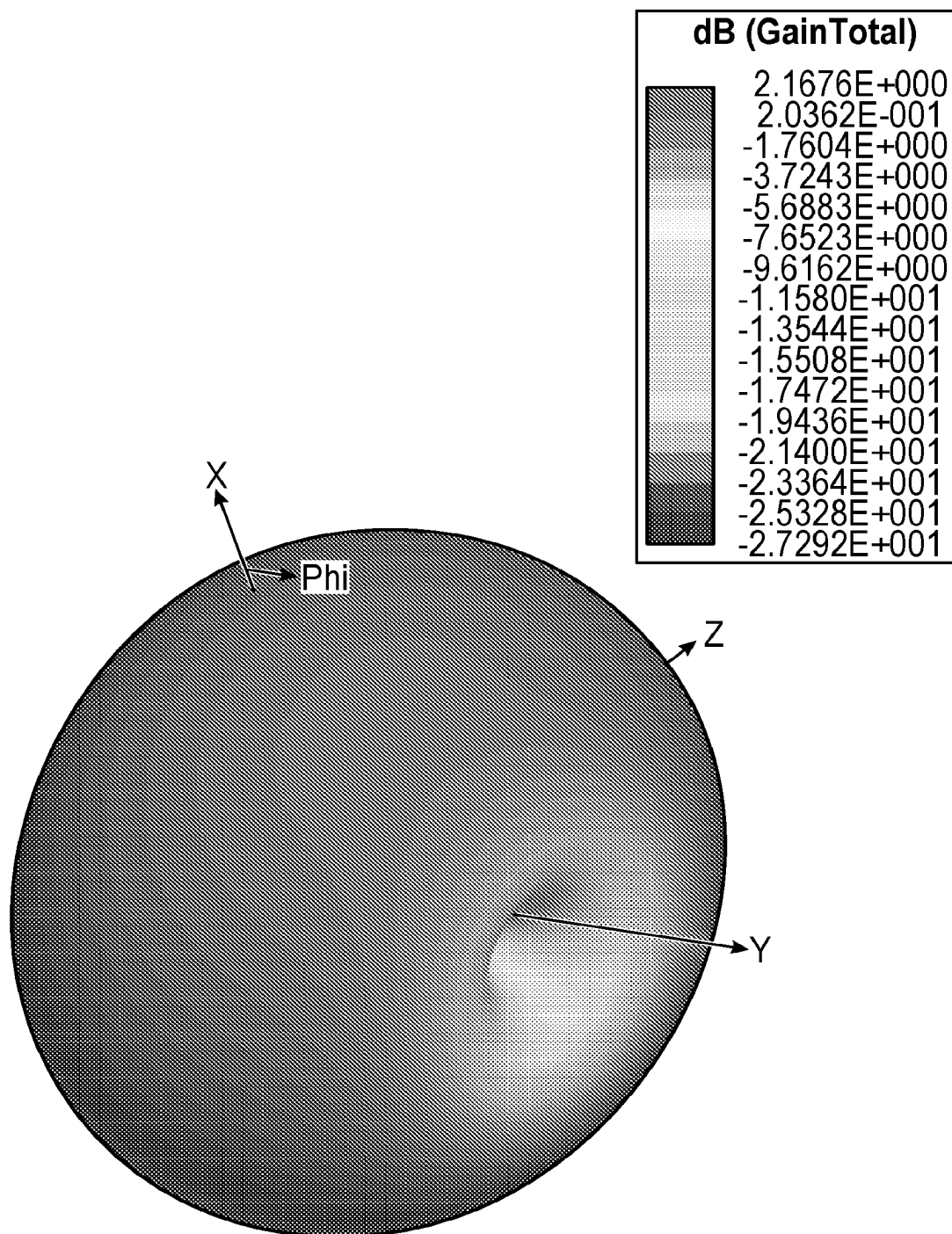


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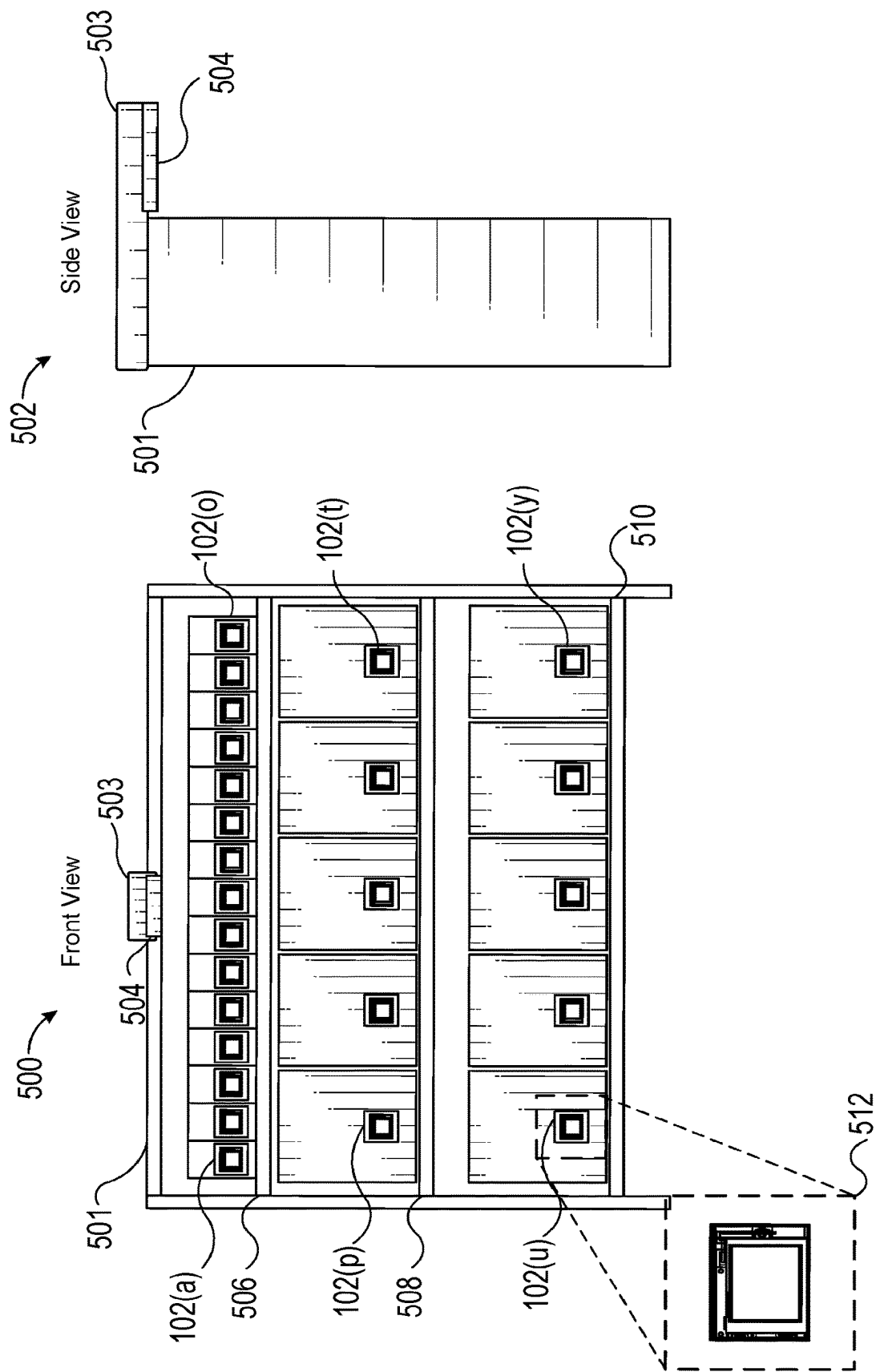


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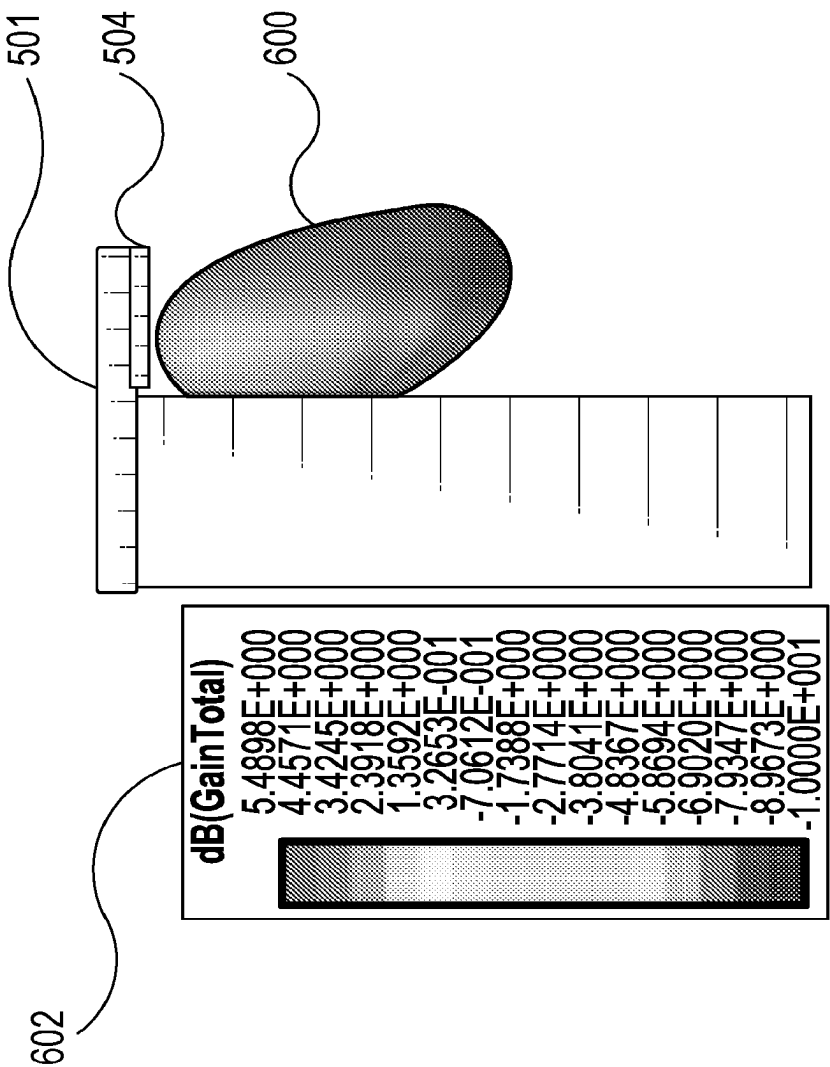


Figure 6

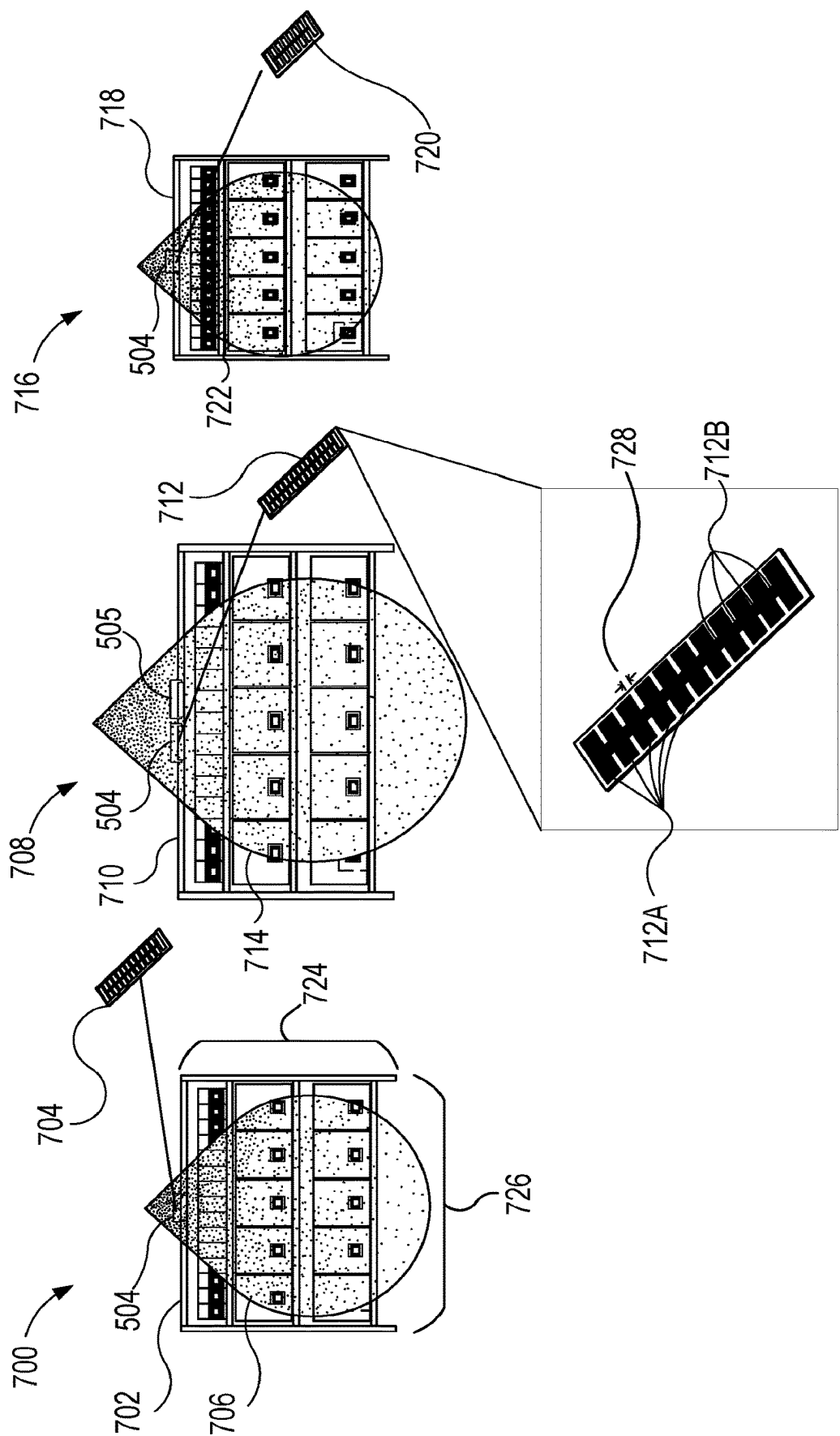


FIG. 7

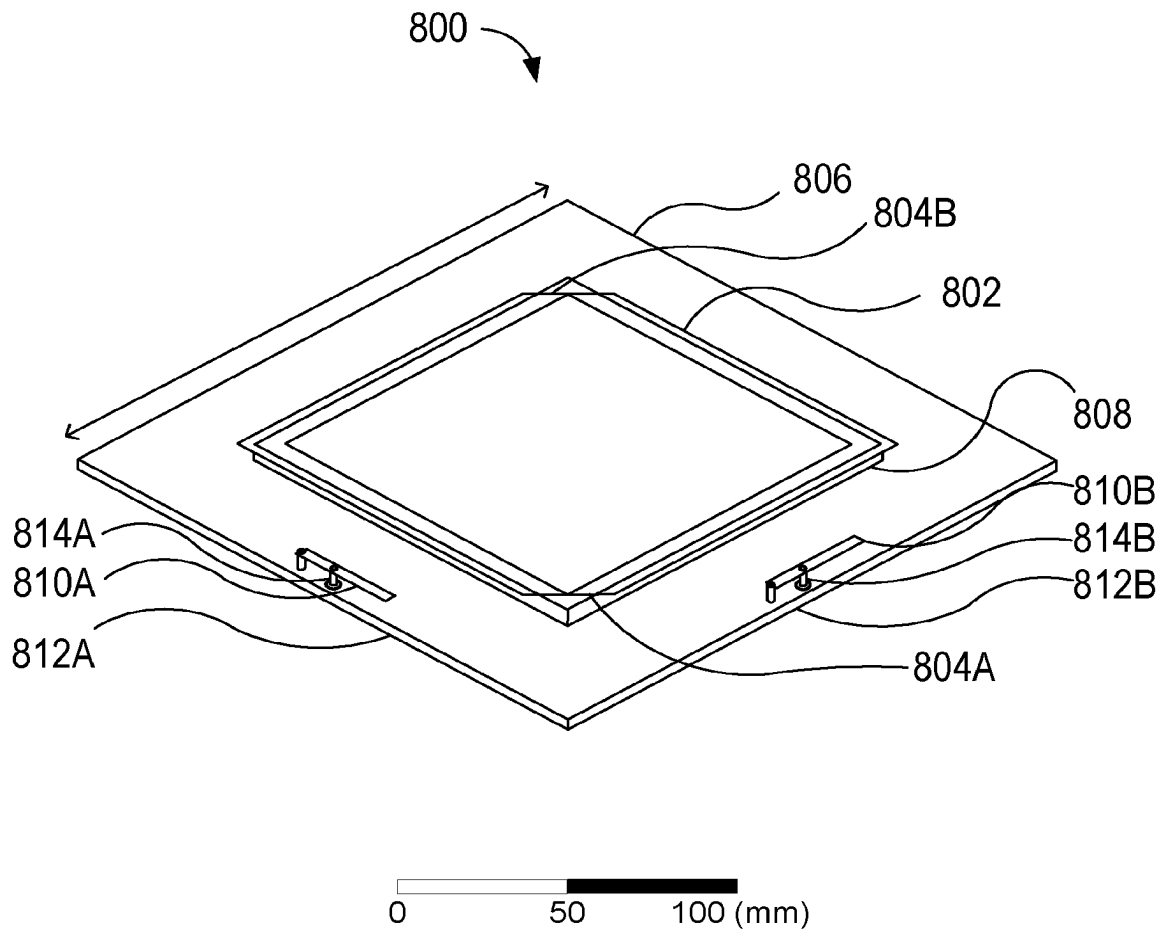


Figure 8

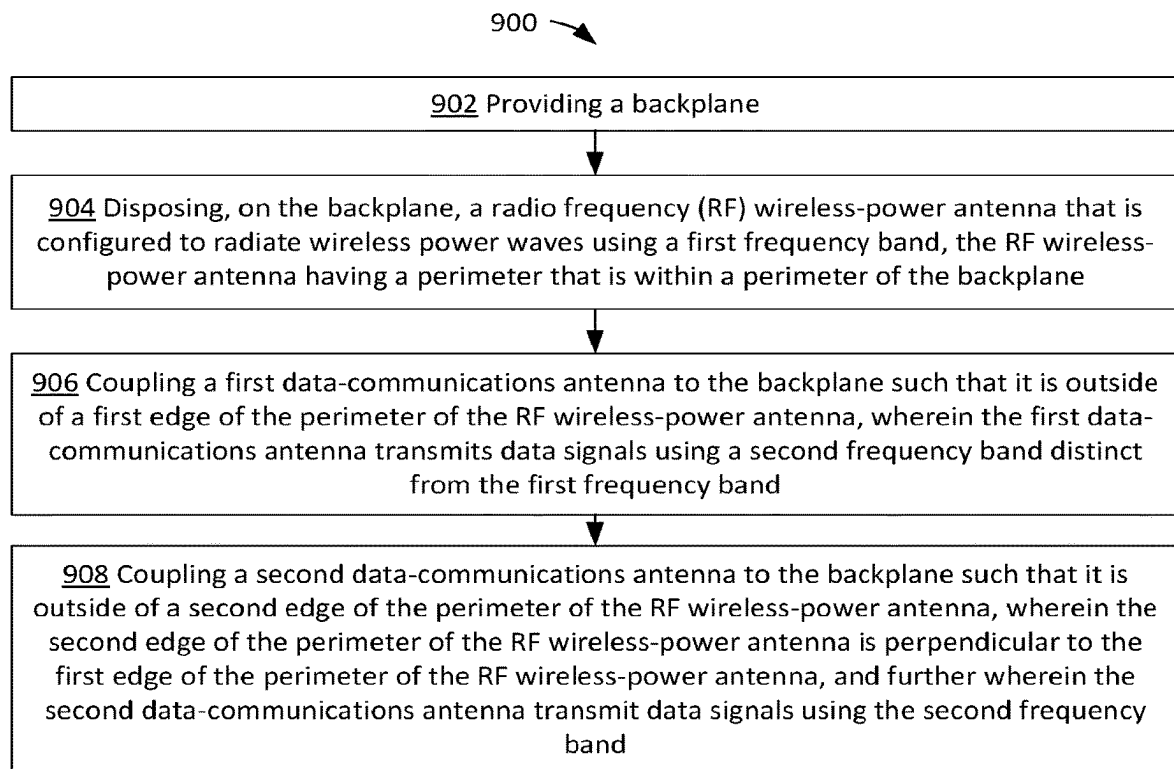


Figure 9

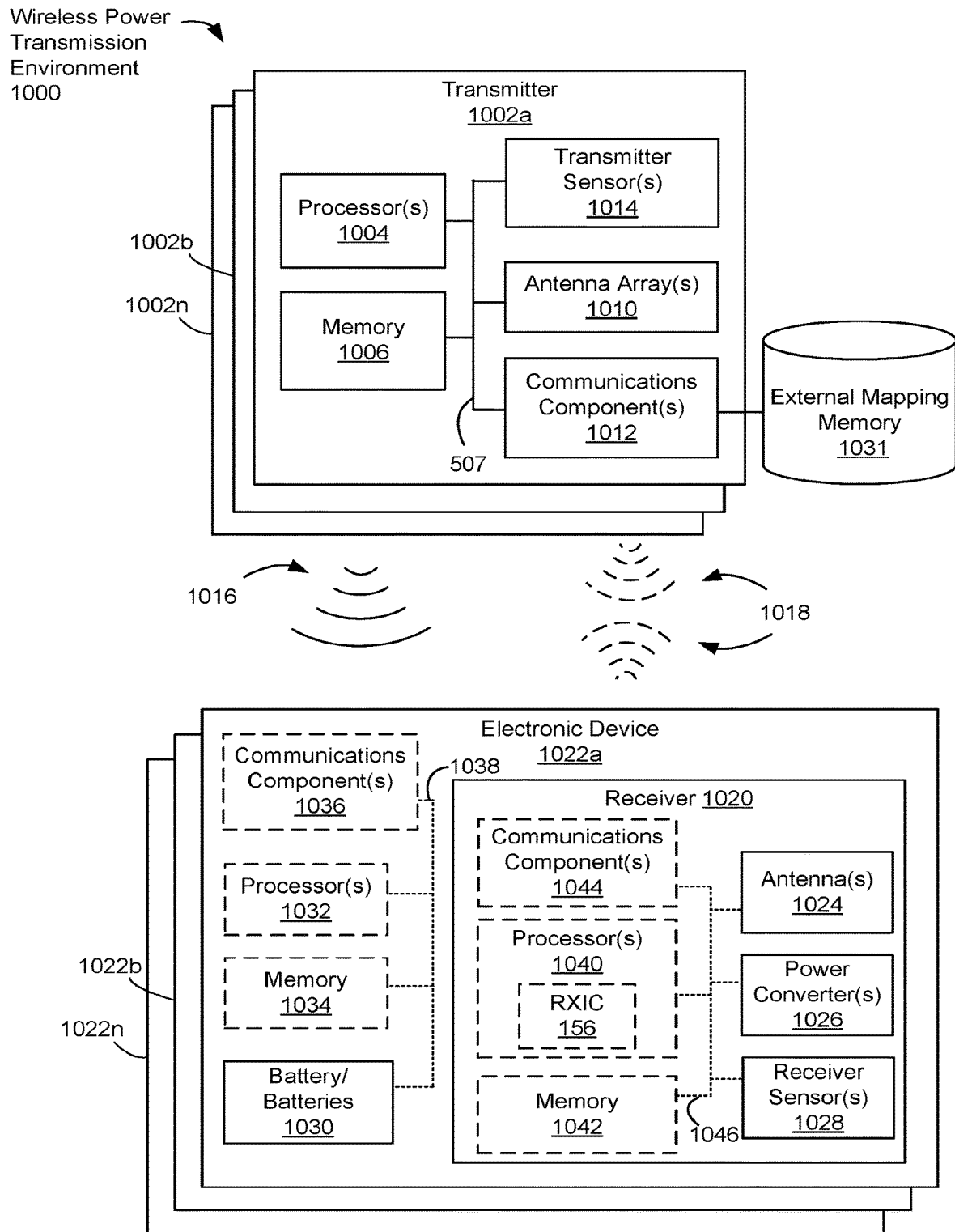


Figure 10

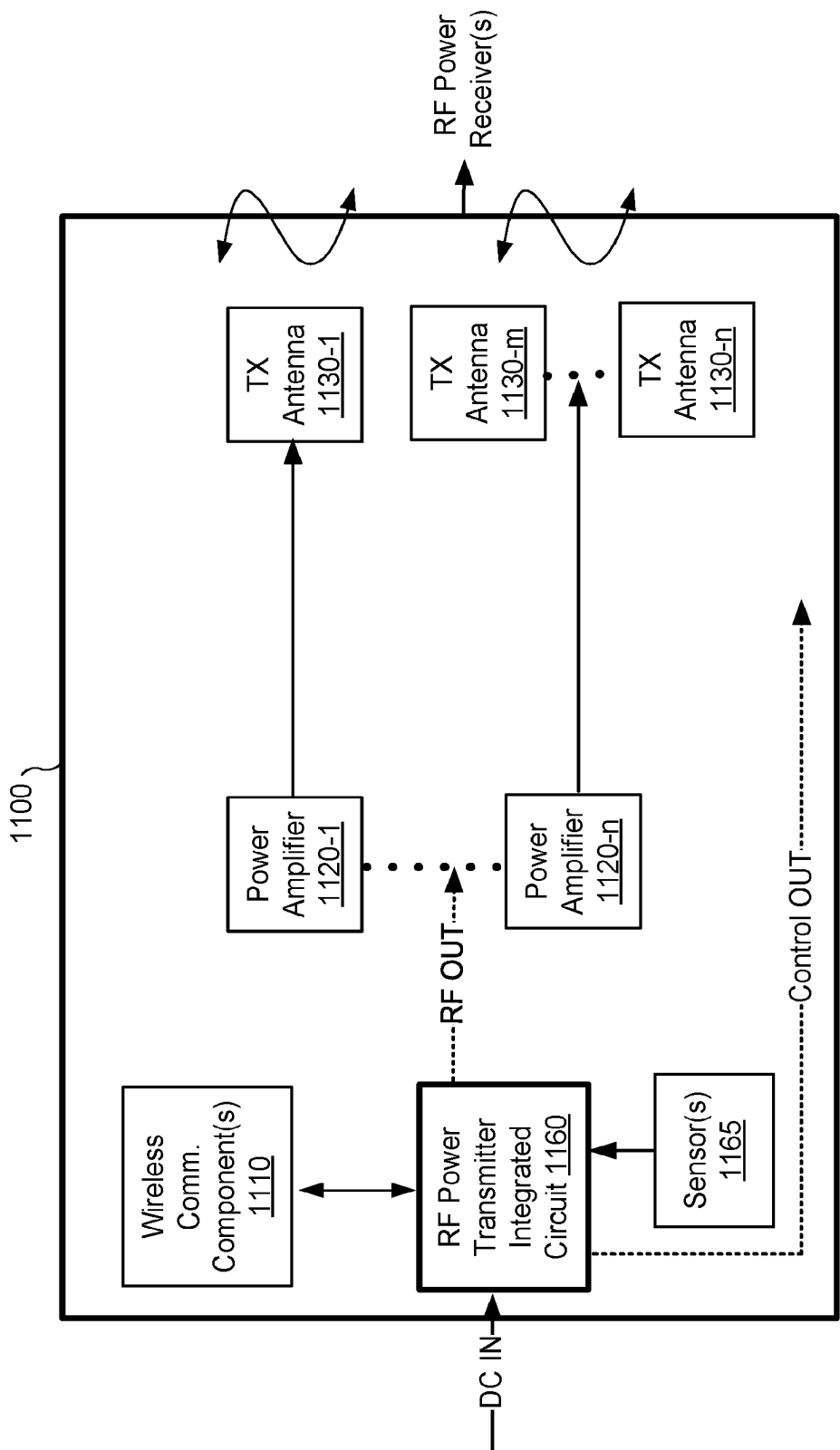


Figure 11A

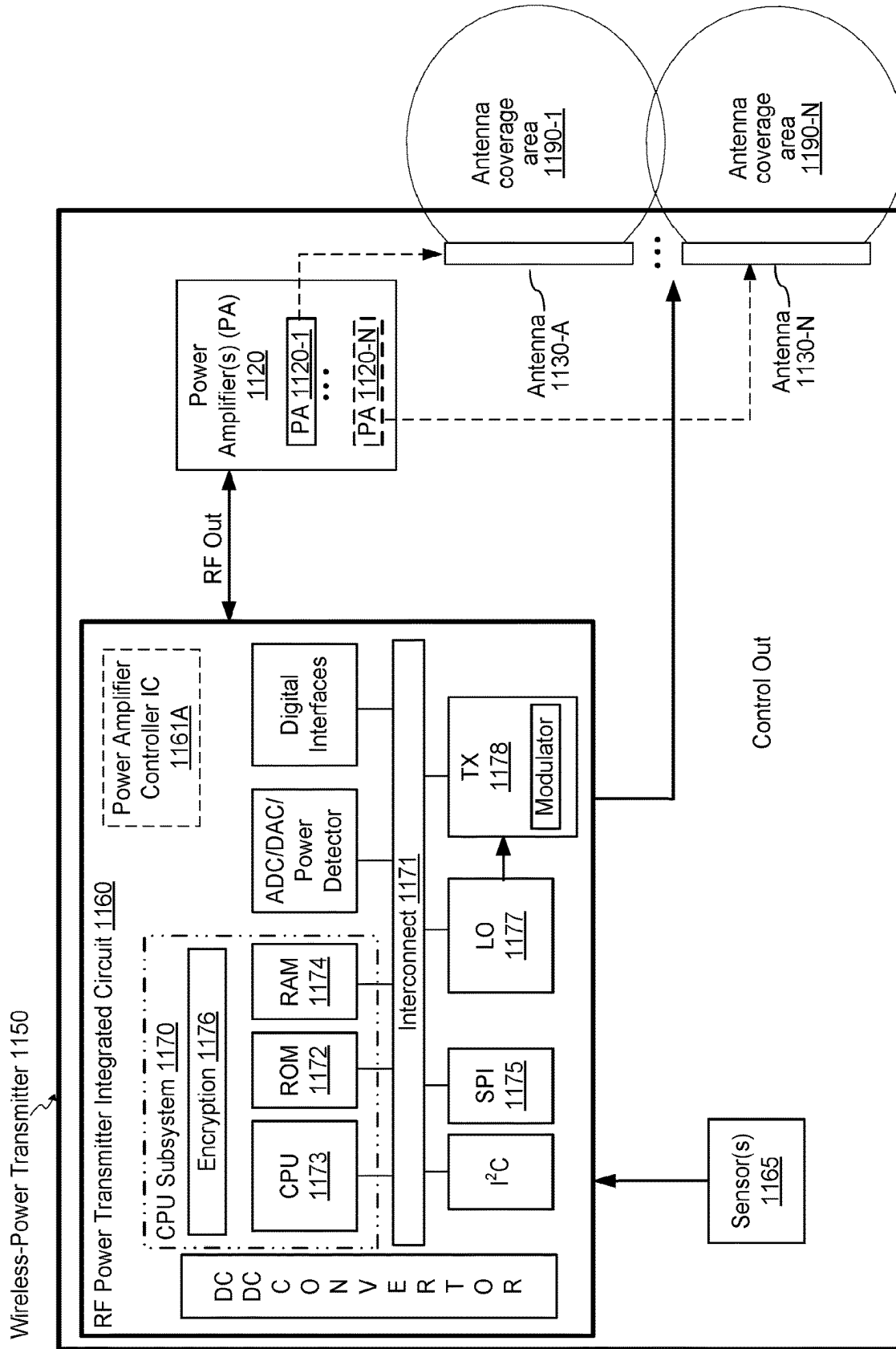


Figure 11B

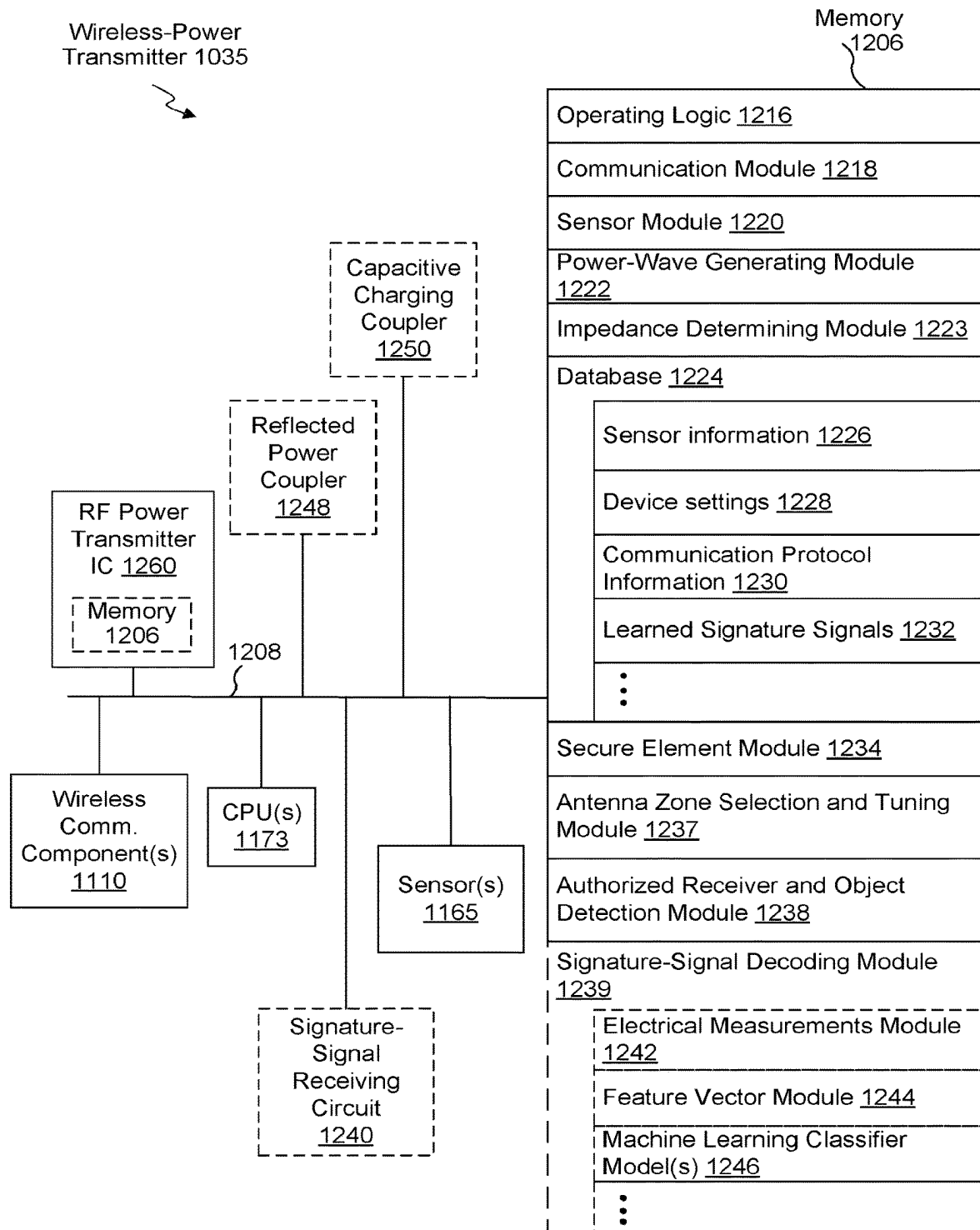


Figure 12

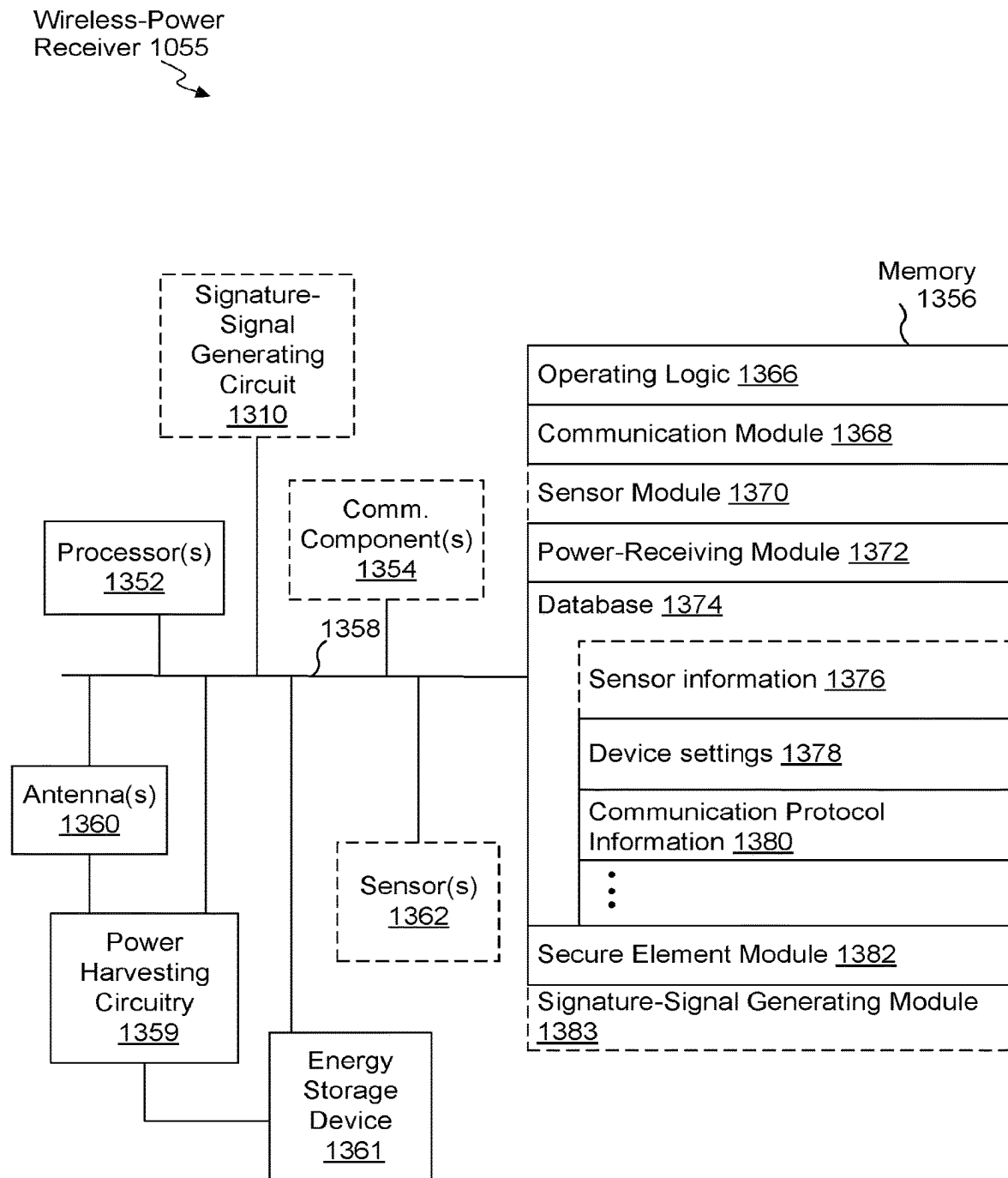


Figure 13

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**SMALL FORM-FACTOR DEVICES WITH
INTEGRATED AND MODULAR
HARVESTING RECEIVERS, AND
SHELVING-MOUNTED WIRELESS-POWER
TRANSMITTERS FOR USE THEREWITH**

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/146,314, filed on Dec. 23, 2022, entitled “Small Form-Factor Devices With Integrated And Modular Harvesting Receivers, And Shelving-Mounted Wireless-Power Transmitters For Use Therewith,” which is application claims priority to U.S. Provisional Application Ser. No. 63/294,555 filed on Dec. 29, 2021, entitled “Systems For Harvesting Radio Frequency Wireless Power Using One Or More Of Shelving-Mounted Wireless-Power Transmitters, Small Form-Factor Devices With Integrated And Modular Harvesting Receivers, And Wireless-Power Transmitters Capable Of Providing Data And Power Signals In Multiple Orientations,” each of which is herein fully incorporated by reference in its respective entirety.

TECHNICAL FIELD

The present disclosure relates generally to systems for harvesting radio-frequency wireless power and, in particular, to systems for harvesting RF wireless power using small-form factor devices with integrated and module harvesting receivers (e.g., the small form-factor devices can be digital price tags or small digital devices used to identify product information in warehousing settings) and shelving-mounted wireless-power transmitters. Some systems also include wireless-power transmitters capable of providing data and power signals in multiple orientations (these additional transmitters can replace or supplement the shelving-mounted wireless-power transmitters).

BACKGROUND

Harvesting energy (e.g., ambient energy or energy transmitted with the intention that it be harvested by a receiver) to charge devices is gaining additional attention.

Some harvesting systems require specific placements and orientations of the harvesting receiver relative to a transmitter to obtain sufficient power to make the system useful. Additionally, these systems can be tailor-made for specific electronic devices and/or charging environments, and thus have limited flexibility. Such devices are therefore poorly suited to changing environments (e.g., being moved around), environments with a variety of differing charging requirements, and struggle with devices that might have various different orientations relative to a transmitting device (both for transmitting and receiving power and data signals).

Some harvesting systems also rely on the use of active beam-forming control techniques that require formation of multiple beams of energy, in which beams are formed to create focused energy in an operational area. To create this focused energy, many existing solutions use beam-forming, e.g., controlling phase and other waveform characteristics to produce constructive and/or destructive interference patterns to focus power beams onto a device at a certain location. Beam-forming typically requires multiple antennas, beam-forming algorithm control circuitry and/or algorithms, and multiple power amplifiers, all of which add complexity to the system, and add to overall system costs.

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As such, it would be desirable to provide systems and methods for wirelessly transmitting and harvesting wireless power that address the above-mentioned drawbacks.

SUMMARY

The wireless-power transmission and harvesting system described herein solves one or more of the problems identified above by making use of one or more of three primary embodiments: (i) an RF harvesting receiver that is configured to receive radiated wireless power in multiple different orientations and which includes modular components that can be easily replaced and which are integrated with a small form-factor device (such as a digital price tag); (ii) shelving-mounted wireless-power transmitters, and (iii) wireless-power transmitters capable of providing data and power signals in multiple orientations. As one example pertaining to the second embodiment, the shelving-mounted transmitter can include a radiating antenna element that can be designed with varying numbers of conductive segments. By adjusting the number of conductive segments (which can be done at manufacture time or which can be done on-the-fly using switches or like structures to activate or disable certain conductive segments), the coverage area of a particular shelving-mounted transmitter can be adjusted. This is convenient for applications where the coverage area is not standard across all areas of an environment (e.g., an environment including multiple shelving units in a grocery store or an environment including storage shelves in a warehouse, and the like) and can fluctuate in different areas of the environment such that different required coverage areas are needed (e.g., a coverage area for multiple different shelving units in a store, or various storage shelves used in large warehouse settings). As one example with respect to the third embodiment, the RF transmitter can be configured to radiate RF signals with a circular polarization while concurrently radiating communications signals in a dual linear fashion (e.g., one data antenna can radiate data signals with a horizontal polarization and another perpendicularly-oriented data antenna can radiate data signals with a vertical polarization). Additionally, integrating multi-orientation data and power signals in a transmitter enables further control of the receiving devices. In one example, where the receiving device is an inventory tag, the electronic device can be updated easily to reflect changes in inventory or price. Examples of these improvements are discussed in detail below, which provide solutions to one or more of the problems discussed earlier.

Additionally, any of the embodiments described herein can utilize transmission techniques that do not require any active beam-forming control (e.g., a single antenna can be utilized with a single power amplifier to allow for wirelessly delivering energy to a harvester device), thereby producing efficient systems with fewer components.

(A1) In accordance with some embodiments, a wireless-power harvester is integrated in a small form-factor device. The wireless-power harvester comprises a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape. The meandering shape includes a predetermined number of turns. The first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and the second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB is configured to operate as a ground plane for the stamped metal antenna. An intermedi-

ate portion is disposed between the first end and the second end of the meandering shape coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more RF power waves, harvested by the stamped metal harvesting antenna, into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device.

(A2) In some embodiments of A1, the small form-factor device is a digital price tag that includes a display powered by the battery. The display and the battery are coupled to the PCB.

(A3) In some embodiments of any of A1-A2, the small form-factor device is a digital thermometer powered by the battery.

(A4) In some embodiments of any of A1-A3, the power-conversion circuitry is on a substrate distinct from the PCB.

(A5) In some embodiments of any of A1-A4, the stamped metal harvesting antenna is quasi omnidirectional.

(A6) In some embodiments of any of A1-A5, the stamped metal harvesting antenna is coupled to an edge of the PCB.

(A7) In some embodiments of any of A1-A6, the stamped metal harvesting antenna is modular, such that it can be interchangeably coupled with the PCB and the power-conversion circuitry at a first point in time, and separately coupled with a different PCB of a different small form-factor device and different power-conversion circuitry at a second point in time that is distinct from the first point in time.

(A8) In some embodiments of any of A1-A7, the stamped metal antenna occupies a first area of the housing of the small form-factor device that is adjacent to a second area of the housing occupied by the PCB, and the first and second areas are non-overlapping.

(A9) In some embodiments of any of A1-A8, the RF power waves have a centering operating frequency of 918 MHz.

(A10) In some embodiments of any of A1-A9, the stamped metal harvesting antenna has a gain of at least 2 dB.

(A11) In some embodiments of any of A1-A10, the predetermined number of turns is two.

(A12) In some embodiments of any of A1-A11, the battery has a capacity of 60 to 100 mAh.

(A13) In some embodiments of any of A1-A12, the stamped metal harvesting antenna has a thickness of 60 mils (about 1.5 mm).

(A14) In some embodiments of any of A1-A13, the stamped metal harvesting antenna has a width of around 1 inch.

(A15) In some embodiments of any of A1-A14, the small form-factor device includes a communications component that is coupled to the PCB, the communications component configured to receive data that allows the small form-factor device to display graphical information.

(A16) In some embodiments of any of A1-A15, the graphical information is displayed using a text-only display of the small form-factor device.

(B1) In another aspect, a small form-factor device, comprises a wireless-power harvester. The wireless-power harvester includes a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape. The meandering shape includes a predetermined number of turns. A first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The

PCB is configured to operate as a ground plane for the stamped metal antenna. An intermediate portion, disposed between the first end and the second end, of the meandering shape is coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device. The wireless-power harvester of the small form-factor device of B1 can be configured in accordance with any of A2-A16. A method of operating the wireless-power harvester of any of claims A1-A16 and or the small form-factor device that includes the harvester is also contemplated, the method operates the harvester to allow it to harvest radio-frequency wireless-power waves, which are then converted to usable energy for power or charging the small form-factor device.

(C1) In another aspect, a shelving-mounted wireless-power-transmitting and receiving system comprises a shelving-mounted wireless-power-transmitting device. The shelving-mounted wireless-power-transmitting device is configured to transmit RF power waves to a wireless-power harvester device that is integrated in a small form-factor device. The wireless-power harvester device integrated in the small form-factor device includes a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape. The meandering shape includes a predetermined number of turns. A first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB configured to operate as a ground plane for the stamped metal antenna. An intermediate portion, disposed between the first end and the second end, of the meandering shape is power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more of the RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device.

(C2) In some embodiments of C1, the shelving-mounted wireless-power-transmitting device comprises a mounting structure that is mountable to a shelving unit, the shelving unit having a predetermined height and a predetermined width. The mounting includes a first end coupled to an outer portion of the shelving unit and a second end opposite the first end extending a predetermined distance away from the outer portion of the shelving unit. The second end includes: a ground plane, a first plurality of conductive segments having a first shape and a first size, each of the first plurality of conductive segments disposed over the ground plane. The second end includes a second plurality of conductive segments having a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size. A respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another. The second end includes the first and second pluralities of conductive segments being disposed over the ground plane to form an antenna that is configured to radiate radio-frequency (RF) wireless power waves towards the shelving unit such that (i) the predetermined height and the predetermined width of the shelving unit are within a coverage area of the RF wireless

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power waves, and (ii) a bottom shelf of the shelving unit receives at least a threshold amount of RF energy from the RF wireless power waves. The harvester of C1 or C2 can be configured in accordance with any of A2-A16, and the wireless-power transmitting device can be configured in accordance with any of D2-D11.

(D1) In another aspect, a shelving-mounted wireless-power-transmitting device comprises a mounting structure that is mountable to a shelving unit, the shelving unit having a predetermined height and a predetermined width. The mounting includes a first end coupled to an outer portion of the shelving unit and a second end opposite the first end extending a predetermined distance away from the outer portion of the shelving unit. The second end includes a ground plane, and a first plurality of conductive segments having a first shape and a first size, a second plurality of conductive segments having a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size. The second end includes a respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another. The second end includes the first and second pluralities of conductive segments being disposed over the ground plane to form an antenna that is configured to radiate radio-frequency (RF) wireless power waves towards the shelving unit such that (i) the predetermined height and the predetermined width of the shelving unit are within a coverage area of the RF wireless power waves, and (ii) a bottom shelf of the shelving unit receives at least a threshold amount of RF energy from the RF wireless power waves.

(D2) In some embodiments of D1, each conductive segment of the first plurality of conductive segments and the second plurality of conductive segments are separated by a predetermined gap.

(D3) In some embodiments of any of D1-D2, the first shape and the second shape is a meandering path that produces a rectangular shape.

(D4) In some embodiments of any of D1-D3, the first size is larger than the second size.

(D5) In some embodiments of any of D1-D4, the first plurality conductive segments and the second plurality of conductive segments each include a predetermined number of two or more conductive segments.

(D6) In some embodiments of any of D1-D5, the outer portion of the shelving unit is a centrally located between two posts of the shelving unit.

(D7) In some embodiments of any of D1-D6, respective numbers of segments in the first and second pluralities of conductive segments are adjustable to allow for creating an altered coverage area for the shelving-mounted wireless-power transmitting device.

(D8) In some embodiments of any of D1-D7, a peak gain of the antenna is at least 5.5 dBi and the wireless power waves are radiated using a center operating frequency of 918 MHz.

(D9) In some embodiments of any of D1-D8, the first plurality of conductive segments and the second plurality of conductive segments are formed using respective stamped copper plates.

(D10) In some embodiments of any of D1-D9, the first plurality of conductive segments includes a first number of conductive segments, the first number of conductive segments selected based on the predetermined height and width of the shelving unit. The second plurality of conductive segments includes a second number of conductive segments,

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the second number also selected based on the predetermined height and width of the shelving unit.

(D11) In some embodiments of any of D1-D10, the shelving-mounted wireless-power-transmitting device further comprises a second mounting structure that is mountable to a second shelving unit, the second shelving unit having a larger width and height than the predetermined height and width of the shelving unit, and the second mounting structure having a ground plane. The shelving-mounted wireless-power-transmitting device further comprises a third plurality of conductive segments with each adjacent conductive segment of the third plurality is separated by a conductive segment of a fourth plurality of conductive segments. The third and fourth pluralities of conductive segments are disposed over the ground plane to form a second antenna that is configured to radiate second RF wireless power waves towards the second shelving unit such that the larger width and height of the second shelving unit is with a second coverage area of the second RF wireless power waves. A third number of conductive segments in the third plurality of conductive segments is larger than the first number of conductive segments and a fourth number of conductive segments in the fourth plurality of conductive segments is larger than the second number of conductive segments, and the second coverage area is larger than the coverage area.

(E1) In another aspect, a wireless-power-transmitting device, includes a backplane that includes a radio frequency (RF) wireless-power antenna that is configured to radiate wireless power waves using a first frequency band. The RF wireless-power antenna has a perimeter that is within a perimeter of the backplane. In some embodiments, the backplane includes a first data-communications antenna configured to transmit data signals using a second frequency band distinct from the first frequency band, and the first data-communications antenna being coupled to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna. The backplane includes a second data-communications antenna configured to transmit data signals using the second frequency band, and the second data-communications antenna is coupled to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna.

(E2) In some embodiments of E1, the wireless-power-transmitting device includes a spacer that is coupled between the RF wireless-power antenna and the backplane such that the RF wireless-power antenna is coupled to the spacer and sits above the backplane.

(E3) In some embodiments of any of E1-E2, the RF wireless-power antenna is circularly polarized such that the wireless power waves using the first frequency band are transmitted having a circular polarization.

(E4) In some embodiments of any of E1-E3, the first data communication antenna has a first polarization and the second communication antenna has a second polarization that is different from the first polarization.

(E5) In some embodiments of any of E1-E4, the first and second edges of the RF wireless-power antenna are separated by a third notched edge that separates the first and second edges and is shorter in length than the first and second edges, such that the RF wireless-power antenna has a generally quadrilateral shape with at least one notched edge removed from a corner of the quadrilateral.

(E6) In some embodiments of any of E1-E5, the RF wireless-power antenna has two notched edges, the two notched edges being symmetrically shaped.

(E7) In some embodiments of any of E1-E6, the first communication antenna is placed in a middle of the first edge of the perimeter of the RF wireless-power antenna, and the second communication antenna is placed in a middle of the second edge of the perimeter of the RF wireless-power antenna.

(E8) In some embodiments of any of E1-E7, the RF wireless-power antenna is a patch antenna.

(E9) In some embodiments of any of E1-E8, the patch antenna is constructed of copper material.

(E10) In some embodiments of any of E1-E9, a peak gain of the RF wireless-power antenna is greater than 8 dBi while the first frequency band is a center frequency band of 918 MHz.

(E11) In some embodiments of any of E1-E10, the wireless-power antenna matching is less than -10 dB.

(E12) In some embodiments of any of E1-E11, the first data-communications antenna and the second data-communications antenna produce a peak gain of 5 dBi while operating at the second frequency band of approximately 2.45 GHz.

(E13) In some embodiments of any of E1-E12, the first data-communications antenna and the second data-communications antenna have a matching of greater than -7 dB.

(E14) In some embodiments of any of E1-E13, the RF frequency wireless power antenna improves the gain of both the first data-communications antenna and the second data-communications antenna as a result of their proximity to the wireless-power antenna.

(E15) In some embodiments of any of E1-E14, the first data-communications antenna and the second data-communications antenna use the RF frequency wireless power antenna and a main ground to provide maximum gain and optimum radiation performance of the first data-communications antenna and the second data-communications antenna.

(E16) In some embodiments of any of E1-E15, the first data-communications antenna and the second data-communications antenna are formed using respective stamped copper plates.

(E17) In some embodiments of any of E1-E16, the first data-communications antenna and the second data-communications antenna are each suspended 0.1 to 0.5 inches from a top surface of the backplane.

(F1) In another aspect, a wireless-power transmitting and receiving system, includes a plurality of wireless power-transmitting devices including any of the shelving-mounted wireless power transmitting devices of D1-D11 or the wireless-power transmitting device of any of E1-E16, or both, a plurality of wireless-power harvesters structured in accordance with any of A1-A16, B1, and C1-C2.

(G1) In another aspect, a method of manufacturing a wireless-power device comprises providing a backplane is performed. The method comprises disposing, on the backplane, a radio frequency (RF) wireless-power antenna that is configured to radiate wireless power waves using a first frequency band, the RF wireless-power antenna having a perimeter that is within a perimeter of the backplane. The method also comprises coupling a first data-communications antenna to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna. The first data-communications antenna transmits data signals using a second frequency band distinct from the first frequency band. The method further comprises coupling a second

data-communications antenna to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna. The second data-communications antenna transmits data signals using the second frequency band.

Note that the various embodiments described above can be combined with any other embodiments described herein. The features and advantages described in the specification are not all inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the present disclosure can be understood in greater detail, a more particular description may be had by reference to the features of various embodiments, some of which are illustrated in the appended drawings. The appended drawings, however, merely illustrate pertinent features of the present disclosure. As one of skill will appreciate upon reading this disclosure in conjunction with the figures, the description may admit or contemplate other features.

FIG. 1 illustrates an example environment (e.g., a warehouse) that includes multiple of the wireless-power-transmitting devices described here and multiple wireless-power receivers that have different orientations but still harvest sufficient power, in accordance with some embodiments.

FIGS. 1A and 1B illustrate different views of a wireless-power harvester integrated in an electronic device, in accordance with some embodiments.

FIG. 2 illustrates an integrated wireless-power harvester enclosed by a housing of a small form-factor electronic device, in accordance with some embodiments.

FIGS. 3A and 3B illustrate different bottom views of an integrated wireless-power harvester, in accordance with some embodiments.

FIG. 4 illustrates a radiation pattern produced by an embodiment of the wireless-power harvester shown in FIGS. 1A and 1B, in accordance with some embodiments.

FIG. 5 illustrates a shelving-mounted wireless power transmitting system for powering and/or charging one or more wireless-power harvesters integrated into small form-factor electronic devices, in accordance with some embodiments.

FIG. 6 illustrates an example cross-section of a gain distribution (emitted radio frequency waves) from a transmitter system placed at the top of a shelving system, in accordance with some embodiments.

FIG. 7 illustrates three example variations of the shelving-mounted wireless power transmitting system for powering and/or charging one or more wireless-power harvesters integrated into small form-factor electronic devices shown in FIGS. 5-6, in accordance with some embodiments.

FIG. 8 illustrates a wireless transmitting device 800 that is configured to emit RF wireless power waves using a first frequency band and data-communication waves using a second frequency band, distinct from the first frequency band, in accordance with some embodiments.

FIG. 9 is a flow diagram showing a method of manufacturing a wireless-power device, in accordance with some embodiments.

FIG. 10 is a block diagram of components of a wireless power transmission environment, in accordance with some embodiments.

FIG. 11A is a block diagram of a wireless-power transmitter, in accordance with some embodiments.

FIG. 11B is a block diagram of another wireless-power transmitter including an RF power transmitter integrated circuit, one or more sensors, one or more antennas, and/or a power amplifier, in accordance with some embodiments.

FIG. 12 is a block diagram illustrating one or more components of a wireless-power transmitter, in accordance with some embodiments.

FIG. 13 is a block diagram illustrating a representative wireless-power receiver, in accordance with some embodiments.

In accordance with common practice, the various features illustrated in the drawings may not be drawn to scale. Accordingly, the dimensions of the various features may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may not depict all of the components of a given system, method or device. Finally, like reference numerals may be used to denote like features throughout the specification and figures.

DETAILED DESCRIPTION

Numerous details are described herein in order to provide a thorough understanding of the example embodiments illustrated in the accompanying drawings. However, some embodiments may be practiced without many of the specific details, and the scope of the claims is only limited by those features and aspects specifically recited in the claims. Furthermore, well-known processes, components, and materials have not been described in exhaustive detail so as not to unnecessarily obscure pertinent aspects of the embodiments described herein.

The transmitter device (also referred to as a wireless-power transmitter device or wireless-power transmitter below) can be an electronic device that includes, or is otherwise associated with, various components and circuits responsible for, e.g., generating and transmitting electromagnetic energy, forming transmission energy within a radiation profile at locations in a transmission field, monitoring the conditions of the transmission field, and adjusting the radiation profile where needed. The radiation profile described herein refers to a distribution of energy field within the transmission range of a transmitter device or an individual antenna (also referred to as a “transmitter”). A receiver (also referred to as a wireless-power receiver) can be an electronic device that comprises at least one antenna, at least one rectifying circuit, and at least one power converter, which may utilize energy transmitted in the transmission field from a transmitter for powering or charging the electronic device.

FIG. 1 illustrates an example set-up of a combination of (i) a plurality of wireless-power harvester devices that are configured to receive radiated wireless power in multiple different orientations and include modular components that can be easily replaced and which are integrated with a small form-factor device (such as a digital price tag); (ii) a plurality of shelving-mounted wireless-power transmitters, and (iii) a wireless-power transmitters capable of providing data and power signals in multiple orientations.

As stated above, FIG. 1 illustrates two types of wireless-power transmitters configured to radiate RF signals to wireless-power harvester devices. Wireless-power transmitter 10 illustrates a first type of wireless-power transmitter that can

be placed on walls, ceilings, or any other supporting structure within a building. The wireless-power transmitter is configured to emit RF wireless-power waves 12 to wireless-power harvester devices (e.g., wireless-power harvester device 14) and is also configured to emit data-communication signals 16A and 16B (unilaterally and bidirectionally) to one or more wireless-power harvester devices (e.g., wireless-power harvester device 14) using communication antennas 18A and 18B (e.g., BLE antennas). Wireless-power harvester device 14, while shown in a certain orientation, can be placed in any other orientation and still receive sufficient power (e.g., enough power to enable the harvester to provide usable power to power or charge an associated small form-factor electronic device, which can be a digital price tag in some embodiments), due in part to the circular polarization of the RF wireless-power waves. Additionally, the wireless-power harvester device 14 can also receive data-communication signals having any orientation, due to the dual linear polarization of the data-communication signals that allows those data-communication signals to be transmitted with both horizontal and vertical polarizations. Further details regarding the first type of wireless-power transmitter are discussed below in reference to FIG. 8. Further details regarding the wireless-power harvester device are discussed in reference to FIGS. 1A-4.

Shelving-mounted wireless-power transmitters 20 and 22 are a second type of wireless-power transmitter. These shelving-mounted wireless-power transmitters can be manufactured or dynamically adjusted to have different numbers of conductive segments in their respective antennas to adjust emission profiles for RF wireless-power waves. FIG. 1 shows shelving-mounted wireless-power transmitter 20 having a first number of conductive segments, which results in emission of RF power waves 24 to wireless-power harvester devices (e.g., wireless-power harvester device 28). Shelving-mounted wireless-power transmitter 22 has a second number of conductive segments, which results in emission of RF power waves 26 to wireless-power harvester devices (e.g., wireless-power harvester device 30). Wireless-power harvester devices 28 and 30, while shown in a certain orientations, can be placed in any other orientation and still receive sufficient power. Additionally, the wireless-power harvester devices 28 and 30 can also receive data-communication signals in any orientation. In some embodiments, it is also possible for the wireless-power harvester devices to harvest additional RF power waves from other nearby wireless-power transmitters (e.g., wireless-power transmitter 10). Further detail of the second type of wireless-power transmitter is discussed in reference to FIG. 5-7. Further detail regarding the wireless-power harvester device is discussed in reference to FIGS. 1A-4.

FIGS. 1A and 1B illustrate different views of a wireless-power receiver (also referred to as a RF harvesting receiver) integrated in a small form-factor electronic device, in accordance with some embodiments. FIG. 1A shows a front view 100 of an electronic device 102 including a wireless-power receiver (e.g., wireless-power receiver 1020; FIG. 10). In some embodiments, the wireless-power receiver is integrated in the electronic device 102. The wireless-power receiver includes a harvester antenna 104 (e.g., a stamped metal harvester antenna, and/or an antennas 1024 described in reference to FIG. 10 below) coupled to a printed circuit board (PCB 108) of the electronic device 102. The electronic device 102 includes a display 106 powered by a power supply (e.g., a battery 1030; FIG. 10) of the electronic device 102. For example, FIG. 1 illustrates display 106 of each respective electronic device showing different information

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(e.g., temperature values for a shelf, quantity values of items for a shelf, and prices of items on a shelf, etc.). The wireless-power receiver is configured to receive wireless power transmitted by a wireless-power transmitter (e.g., transmitter **1002** of FIG. **10**) such that electromagnetic energy is wirelessly transferred from the wireless-power transmitter to the wireless-power receiver, as described below. FIG. **1B** shows a bottom view **150** of an electronic device **102** including a wireless-power receiver (e.g., wireless-power receiver **1020**; FIG. **10**). The bottom view **150** of the electronic device **102** shows the harvester antenna **104** coupled to the PCB **108** of the electronic device **102** and a receiver integrated circuit **156**.

In some embodiments, the harvester antenna **104** is a stamped metal antenna that has a meandering shape. The meandering shape includes a predetermined number of turns or curves. In some embodiments, the predetermined number of turns is at least two. The harvester antenna **104** is quasi omnidirectional and configured to receive wireless power with any polarization. In some embodiments, a position of the harvester antenna **104** within the electronic device **102** is based, in part, on a size and/or a shape of a housing enclosing the wireless-power receiver (e.g., housing of electronic device **102**). In some embodiments, the harvester antenna **104** has a thickness of 60 mils~1.5 mm (+/-0.5 mm) and a width of 25 mm (+/-15 mm), which is indicated by reference numerals **103** and **105**, respectively, in FIG. **3A**. In some embodiments, the length of the harvester antenna **104** is based, in part, on the size of the electronic device **102**, the position of the harvester antenna **104** within the electronic device **102**, the number of turns included in the harvester antenna **104**, and the shape of the electronic device **102**.

In some embodiments, a first end **160** of the harvester antenna **104** (e.g., the meandering shaped antenna) is a free end configured to receive one or more radio frequency (RF) power waves. In some embodiments, the harvester antenna **104** is configured to receive the RF power waves at a frequency of 918 MHz. In some embodiments, the receiving antenna **104** has a gain of at least 2 dB (shown and described below in reference to FIG. **4**). In some embodiments, the harvester antenna **104** is configured to harvest RF energy transmitted in the environment from background sources other than a dedicated transmitter.

In some embodiments, an intermediate portion **164** of the receiving antenna **104** is coupled to a receiver integrated circuit **156** (e.g., receiver integrated circuit **156**). As discussed below, the receiver integrated circuit **156** is configured to convert one or more RF power waves received by the first end of the meandering shape into usable energy for charging a battery of the electronic device **102** or for powering the electronic device **102**. In some embodiments, the receiver integrated circuit **156** is an integral part of the PCB **108**. Alternatively, in some embodiments, the receiver integrated circuit **156** is on a substrate distinct from the PCB **108**.

In some embodiments, a second end **162** of the harvester antenna **104** is coupled to the PCB **108** of the electronic device **102**—more specifically, an edge **154** of the PCB **108**. In this way, the PCB **108** is configured to operate as a reference ground plane of the harvester antenna **104**. The second end **162** of the harvester antenna **104** can be coupled to any edge portion of the PCB **108**. In some embodiments, the edge portion of the PCB **108** at which the harvester antenna **104** is coupled is based, in part, on a size and/or a shape of a housing enclosing the wireless-power receiver

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(e.g., housing of electronic device **102**) such that the first end **160** of the harvester antenna **104** is able to receive wireless power.

In some embodiments, the electronic device **102** is a digital price tag including a display **106**. The electronic device **102** can be any small form factor device. For example, the electronic device **102** can be a digital price tag, a digital timer, a digital thermometer, a digital scale, an aspect of a smart shopping cart, a tablet, a controller, or other components used to manage inventory and pricing in supermarket or warehouse contexts, etc. In some embodiments, the electronic device **102** includes a communications component (e.g., communications component **1036** and/or **1044**; FIG. **10**) configured to communicatively couple to one or more electronic devices and/or wireless-power transmitters. For example, the electronic device **102** can be located within a supermarket and communicatively couple with a plurality of wireless-power transmitters **502** positioned within the supermarket and/or a computer network within the supermarket. In some embodiments, the electronic device **102** transmits and/or receives data to the one or more communicatively coupled electronic devices and/or wireless-power transmitters **502**. The data can include charging information such as battery life, amount of power received, charge requests, etc. In some embodiments, the data can include electronic device **102** specific data. For example, a digital price tag can transmit data regarding the number of objects on a shelf, object price, total stock of an object, discount or sales related to the object, etc. Similarly, the digital price tag can receive data updating a price or other information of an object.

In some embodiments, the electronic device **102** can be located within a supermarket and communicatively couple with a plurality of wireless-power transmitters **502** positioned within a warehouse for actively updating inventory levels. A plurality of electronic devices **102** and a plurality of wireless-power transmitters **502** may be positioned around the warehouse. Additionally, the plurality of electronic devices **102** and a plurality of wireless-power transmitters **502** in the warehouse can be configured to communicate with a network to update the displays of the electronic devices to reflect inventory levels. In some embodiments, the electronic devices **102** are in communication with the network. In some embodiments, plurality of electronic devices **102** communicate with the plurality of wireless-power transmitters **502**, and the plurality of wireless-power transmitters **502** communicate with the network.

The harvester antenna **104** is interchangeable. More specifically, the harvester antenna **104** is configured such that it can be positioned and/or repositioned along any edge of the PCB **108** of an electronic device **102**. The harvester antenna **104** can be designed to accommodate different configurations of the electronic device **102**. For example, the receiving antenna **104** can be coupled to different electronic devices including different PCB layouts, distinct components, and/or distinct housings. The harvester antenna **104** can be retrofitted into existing electronic devices **102** and or incorporated into an electronic device **102** during manufacturing. The harvester antenna **104** is inexpensive and easy to manufacture, which increases the number of electronic devices **102** in which the harvester antenna **104** can be integrated. This increases the availability of wireless power to an electronic device **102**.

FIG. **2** illustrates an integrated wireless-power receiver enclosed by an electronic device, in accordance with some embodiments.

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FIGS. 3A and 3B illustrate different perspective views of the integrated wireless-power receiver discussed in FIGS. 1A-1B and 2, in accordance with some embodiments. This figures show that the integrated wireless-power receiver fits within the housing of the small form-factor device, such that no changes to dimensions of the small form-factor device need be made to account the integrate wireless-power receiver. FIG. 3A illustrates an optional battery 148 (e.g., battery 1030 shown in FIG. 10) placed within the housing of the small form-factor device.

FIG. 4 illustrates a radiation pattern produced by an embodiment of a wireless-power receiver shown in FIGS. 1A and 1B. As shown, the radiation pattern 400 is substantially omnidirectional. More specifically, the wireless-power receiver has a 2 dB gain in the X, Y, and Z directions. In some embodiments, the wireless-power receiver is able to achieve an efficiency of approximately 94%.

FIG. 5 illustrates a shelving-mounted wireless-power transmitting system for powering and/or charging one or more wireless-power receiver integrated into an electronic device (e.g., electronic device 102 in FIGS. 1A and 1B), in accordance with some embodiments. In particular, FIG. 5 shows a front view 500 of a shelving system 501 (e.g., shelving rack) and a side view 502 of a shelving system 501. Front view 500 of a shelving system 501 illustrates a shelving-mounted wireless-power transmitter system 504 placed at the top of the shelving system 501 (e.g., shelving-mounted wireless-power transmitter system 504 is placed above the one or more wireless-power receivers) that is configured to project radio frequency (RF) power waves to the one or more wireless-power receivers placed on the different shelves of the shelving system. In some embodiments, the transmitter system is mounted to the shelving system (e.g., shelving unit) via a mounting structure (e.g., mounting structure 503). The front view 500 of the shelving system 501 also shows a plurality of electronic device(s) 102 placed on each shelf of the shelving system. The upper-most shelf 506 has a first plurality of electronic device(s) (e.g., 102(a) through 102(o)) placed on the upper-most shelf. In some embodiments, the first plurality of electronic device(s) are attached to items that are place on the upper most shelf. In some embodiments, the first plurality of electronic device(s) are placed directly (e.g., placed on an outward edge of the upper-most shelf) on the shelves at locations corresponding to items resting on the shelf. Front view 500 of the shelving system 501 also illustrates a middle shelf 508 that includes a second plurality of electronic device(s) 102 (e.g., 102(p) through 102(t)) placed on it, and a bottom shelf 510 that includes a third plurality of electronic device(s) 102 (e.g., 102(u) through 102(y)). Front view 500 of the shelving system 501 also shows an expanded electronic device 512, which corresponds to the electronic devices discussed in reference to FIG. 1A through FIG. 4. FIG. 5 also illustrates a side view 502 of a shelving system 501 that shows the placement of the shelving-mounted wireless-power transmitter system 504 relative to the shelving system 501.

FIG. 6 illustrates an example cross-section 600 of the gain distribution (emitted radio frequency waves) from shelving-mounted wireless-power transmitter system 504 placed at the top of the shelving system 501, in accordance with some embodiments. FIG. 6 also illustrates a table 602 for interpreting the gradation of the cross-section 600 of the gain.

FIG. 7 illustrates three example variations of the shelving-mounted wireless power transmitting system for powering and/or charging one or more wireless-power receiver integrated into an electronic device shown in FIGS. 5-6, in accordance with some embodiments. FIG. 7 illustrates that

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the shelving-mounted wireless-power transmitter system 504 can be adjusted to account for varying sizes of shelving systems. In some embodiments, this adjustment is at least partially done by altering the number of conductive segments to apply coverage for varying shelving systems having different widths, depths, heights. In some embodiments, the number of conductive segments are increased (e.g., more conductive segments are added in series) in order to increase the coverage area (e.g., horizontal coverage area) of the shelving-mounted wireless-power transmitter system 504.

FIG. 7 shows a first example shelving-mounted wireless power transmitting system 700 that is configured to radiate radiofrequency (RF) wireless power waves towards the shelving unit of a first size 702 with a predetermined height 724 and a predetermined width 726. In order to radiate RF wireless power waves to all the receivers on the shelving unit of a first size 702, a first number of conductive segments 704 are included in the shelving-mounted wireless-power transmitter system 504. As a result of the first number of conductive segments being included in the shelving-mounted wireless-power transmitter system 504, a first RF coverage area 706 is produced.

FIG. 7 also shows a second example shelving-mounted wireless power transmitting system 708 that is configured to radiate RF wireless power waves towards the shelving unit of a second size 710 (e.g., larger than the shelving unit of the first size 702). In order to radiate RF wireless power waves to all the receivers on the shelving unit of a second size 710, a second number of conductive segments 712 are included in the shelving-mounted wireless-power transmitter system 504 (e.g., greater in number than the first number of conductive segments 704). In some embodiments, two different pluralities of conductive segments are included, illustrated as a first plurality of conductive segments 712A and a second plurality of conductive segments 712B separating adjacent respective segments of the first plurality from one another by a predetermined gap distance 728. In some embodiments, an additional shelving-mounted wireless-power transmitter system 505 can be mounted to the shelving unit. As a result of the second number of conductive segments being included in the shelving-mounted wireless-power transmitter system 504, a second RF coverage area 714 is produced (e.g., covering more area than the first RF coverage area 706).

FIG. 7 shows a third example shelving-mounted wireless power transmitting system 716 that is configured to radiate RF wireless power waves towards the shelving unit of a third size 718 (e.g., smaller than the shelving unit of the first size 702 and the shelving unit of the second size 710). In order to radiate RF wireless power waves to all the receivers on the shelving unit of a third size 718, a third number of conductive segments 720 are included in the shelving-mounted wireless-power transmitter system 504 (e.g., lesser in number than the first number of conductive segments 704 and the second number of conductive segments 712). As a result of the third number of conductive segments being included in the shelving-mounted wireless-power transmitter system 504, a third RF coverage area 722 is produced (e.g., covering less area than the first RF coverage area 706 and second RF coverage area 714).

FIG. 8 illustrates a wireless transmitting device 800 that is configured to emit RF wireless power waves using a first frequency band and communication waves using a second frequency band, distinct from the first frequency band, in accordance with some embodiments. FIG. 8 illustrates a RF wireless-power antenna 802 that is configured to radiate wireless power waves using a first frequency band. In some

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embodiments, the RF wireless-power antenna **802** is configured to emit power waves in a circular polarized manner. In some embodiments, the RF wireless-power antenna **802** is notched on opposing corners (e.g., a first notch **804A** and a second notch **804B**) of the wireless-power antenna **802**. The opposing notches help aid in the circular polarization of the RF waves emitted by the RF wireless-power antenna **802**. In some embodiments, the RF wireless-power antenna **802** is coupled to a backplane **806**. In some embodiments, the RF wireless-power antenna **802** is coupled to a spacer element **808** that is coupled to the backplane **806**.

FIG. **8** also illustrates that the wireless transmitting device **800** includes two data communication antennas (e.g., a first data communication antenna **810A** and a second data communication antenna **810B**) that are configured to emit communication waves using a second frequency band. The first data communication antenna **810A** and the second data communication antenna **810B** are placed perpendicular to one another on adjacent sides of the backplane **806**. In some embodiments, the first data communication antenna **810A** is placed in the middle of the first edge **812A** and the second data communication antenna **810B** is placed in the middle of the second edge **812B**. In some embodiments, the first data communication antenna **810A** and the second data communication antenna **810B** are placed around the periphery of the RF wireless-power antenna **802**. In some embodiments, the first data communication antenna **810A** is suspended by at least one post (e.g., post **814A**) and the second data communication antenna **810B** is suspended by at least one post (e.g., post **814B**). In some embodiments, at least one post acts as a connection to the ground plane.

FIG. **9** is a flow diagram showing a method of manufacturing a wireless-power device, in accordance with some embodiments. Operations (e.g., steps) of the method **900** may be performed by a manufacturer and/or a manufacturing system. In some embodiments, manufacturing a wireless-power device comprises providing (**902**) a backplane (e.g., a substrate with no computing components). For example, a substrate with no computing components, as shown as backplane **806** in FIG. **8**.

Manufacturing a wireless-power device comprises disposing (**904**), on the backplane, a radio frequency (RF) wireless-power antenna that is configured to radiate wireless power waves using a first frequency band, the RF wireless-power antenna having a perimeter that is within a perimeter of the backplane (e.g., FIG. **8** illustrates that the RF wireless-power antenna **802** having a perimeter within a perimeter of the backplane **806**).

Manufacturing a wireless-power device comprises coupling (**906**) a first data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna. The first data-communications antenna transmits data signals using a second frequency band distinct from the first frequency band (e.g., FIG. **8** illustrates first data communication antenna **810A** coupled to the backplane **806**).

Manufacturing a wireless-power device comprises coupling (**908**) a second data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna. The second data-communications antenna transmit data signals using the second frequency band (e.g., FIG. **8** illustrates

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second data communication antenna **810B** coupled to the backplane **806** perpendicular to the first data communication antenna **810A**).

FIG. **10** is a block diagram of components of wireless power transmission environment **1000**, in accordance with some embodiments. Wireless power transmission environment **1000** includes, for example, transmitters **1002** (e.g., transmitters **1002a**, **1002b** . . . **1002n**) (also referred to as wireless-power-transmitting device and shelving-mounted wireless-power-transmitting device) and one or more receivers **1020** (e.g., receivers **1020a**, **1020b** . . . **1020n**) (also referred to as wireless-power receiver and RF harvesting receiver). In some embodiments, each respective wireless power transmission environment **1000** includes a number of receivers **1020**, each of which is associated with a respective electronic device **1022**. In some instances, the transmitter **1002** is referred to herein as a “wireless-power-transmitting device” or a “wireless power transmitter.” Additionally, in some instances, the receiver **1020** is referred to herein as a “wireless-power-receiving device” or a “wireless-power receiver.”

An example transmitter **1002** (e.g., transmitter **1002a**) includes, for example, one or more processor(s) **1004**, a memory **1006**, one or more antenna arrays **1010**, one or more communications components **1012** (also referred to herein as a “wireless communications radio,” a “communications radio” or simply a “radio”), and/or one or more transmitter sensors **1014**. In some embodiments, these components are interconnected by way of a communications bus **1007**.

In some embodiments, a single processor **1004** (e.g., processor **1004** of transmitter **1002a**) executes software modules for controlling multiple transmitters **1002** (e.g., transmitters **1002b** . . . **1002n**). In some embodiments, a single transmitter **1002** (e.g., transmitter **1002a**) includes multiple processors **1004**, such as one or more transmitter processors (configured to, e.g., control transmission of signals **1016** by antenna array **1010**), one or more communications component processors (configured to, e.g., control communications transmitted by communications component **1012** and/or receive communications by way of communications component **1012**) and/or one or more sensor processors (configured to, e.g., control operation of transmitter sensor **1014** and/or receive output from transmitter sensor **1014**).

The wireless-power receiver **1020** receives power transmission signals **1016** and/or data-communication signals **1018** transmitted by transmitters **1002**. In some embodiments, the receiver **1020** includes one or more antennas **1024** (e.g., an antenna array including multiple antenna elements), power converter **1026**, receiver sensor **1028**, and/or other components or circuitry (e.g., processor(s) **1040**, memory **1042**, and/or communication component(s) **1044**). In some embodiments, these components are interconnected by way of a communications bus **1046**. References to these components of receiver **1020** cover embodiments in which one or more of these components (and combinations thereof) are included.

The receiver **1020** converts energy from received signals **1016** (also referred to herein as RF power transmission signals, or simply, RF signals, RF waves, power waves, or power transmission signals) into electrical energy to power and/or charge electronic device **1022**. For example, the receiver **1020** uses the power converter **1026** to convert energy derived from power waves **1016** to alternating current (AC) electricity or direct current (DC) electricity to power and/or charge the electronic device **1022**. Non-lim-

iting examples of the power converter **1026** include rectifiers, rectifying circuits, voltage conditioners, among suitable circuitry and devices.

In some embodiments, the receiver **1020** is a standalone device that is detachably coupled to one or more electronic devices **1022**. For example, the electronic device **1022** has processor(s) **1032** for controlling one or more functions of the electronic device **1022**, and the receiver **1020** has processor(s) **1040** for controlling one or more functions of the receiver **1020**.

In some embodiments, the receiver **1020** is a component of the electronic device **1022**. For example, processors **1032** control functions of the electronic device **1022** and the receiver **1020**. In addition, in some embodiments, the receiver **1020** includes one or more processors **1040**, which communicates with processors **1032** of the electronic device **1022**.

In some embodiments, the electronic device **1022** includes one or more processors **1032**, memory **1034**, one or more communication components **1036**, and/or one or more batteries **1030**. In some embodiments, these components are interconnected by way of a communications bus **1038**. In some embodiments, communications between electronic device **1022** and receiver **1020** occur via communications component(s) **1036** and/or **1044**. In some embodiments, communications between the electronic device **1022** and the receiver **1020** occur via a wired connection between communications bus **1038** and communications bus **1046**. In some embodiments, the electronic device **1022** and the receiver **1020** share a single communications bus.

In some embodiments, the receiver **1020** receives one or more power waves **1016** directly from the transmitter **1002** (e.g., via one or more antennas **1024**). In some embodiments, the receiver **1020** harvests power waves from one or more pockets of energy created by one or more power waves **1016** transmitted by the transmitter **1002**. In some embodiments, the transmitter **1002** is a near-field transmitter that transmits the one or more power waves **1016** within a near-field distance (e.g., less than approximately six inches away from the transmitter **1002**). In other embodiments, the transmitter **1002** is a far-field transmitter that transmits the one or more power waves **1016** within a far-field distance (e.g., more than approximately six inches away from the transmitter **1002**).

After the power waves **1016** are received and/or energy is harvested from them, circuitry (e.g., integrated circuits, amplifiers, rectifiers, and/or voltage conditioner) of the receiver **1020** converts the energy of the power waves to usable power (i.e., electricity), which powers the electronic device **1022** and/or is stored to battery **1030** of the electronic device **1022**. In some embodiments, a rectifying circuit of the receiver **1020** translates the electrical energy from AC to DC for use by the electronic device **1022**. In some embodiments, a voltage conditioning circuit increases or decreases the voltage of the electrical energy as required by the electronic device **1022**. In some embodiments, an electrical relay conveys electrical energy from the receiver **1020** to the electronic device **1022**.

In some embodiments, the electronic device **1022** obtains power from multiple transmitters **1002** and/or using multiple receivers **1020**. In some embodiments, the wireless power transmission environment **1000** includes a plurality of electronic devices **1022**, each having at least one respective receiver **1020** that is used to harvest power waves from the transmitters **1002** into power for charging the electronic devices **1022**.

In some embodiments, the one or more transmitters **1002** adjust values of one or more characteristics (e.g., waveform characteristics, such as phase, gain, direction, amplitude, polarization, and/or frequency) of power waves **1016**. For example, a transmitter **1002** selects a subset of one or more antenna elements of antenna array **1010** to initiate transmission of power waves **1016**, cease transmission of power waves **1016**, and/or adjust values of one or more characteristics used to transmit power waves **1016**. In some embodiments, the one or more transmitters **1002** adjust power waves **1016** such that trajectories of power waves **1016** converge at a predetermined location within a transmission field (e.g., a location or region in space), resulting in controlled constructive or destructive interference patterns. The transmitter **1002** may adjust values of one or more characteristics for transmitting the power waves **1016** to account for changes at the wireless-power receiver that may negatively impact transmission of the power waves **1016**.

Note that, in some embodiments, the transmitter **1002** utilizes beamforming techniques to wirelessly transfer power to a receiver **1020**, while in other embodiments, the transmitter **1002** does not utilize beamforming techniques to wirelessly transfer power to a receiver **1020** (e.g., in circumstances in which no beamforming techniques are used, the transmitter controller IC **1060** discussed below might be designed without any circuitry to allow for use of beamforming techniques, or that circuitry may be present, but might be deactivated to eliminate any beamforming control capability).

In some conventional systems, a viable RF power level can be provided to an optional beam-forming integrated circuit (IC) (not shown), which then provides phase-shifted signals to one or more power amplifiers. In such conventional systems, the optional beam-forming IC is used to ensure that power transmission signals sent using two or more antennas wireless-power receivers are transmitted with appropriate characteristics (e.g., phases) to ensure that power transmitted to the particular wireless-power receiver is maximized (e.g., the power transmission signals arrive in phase at the particular wireless-power receiver). The embodiments herein, however, do not always require the use of a beam-forming integrated circuit. In certain embodiments, such a beam-forming integrated circuit (and/or associated algorithm) can be included in the system, but is disabled (or is not included in the system at all) and is not used in conjunction with wirelessly-transmitted energy to receiver devices.

In some embodiments, the transmitter controller IC **1060** provides the viable RF power level directly to the one or more power amplifiers and does not use any beam-forming capabilities (e.g., bypasses/disables a beam-forming IC and/or any associated algorithms if phase-shifting is not required, such as when only a single antenna **1024** is used to transmit power transmission signals to a wireless-power receiver). In some embodiments, only a single antenna is utilized with a single power amplifier (e.g., each of the transmitter devices **10**, **20**, **22**, FIG. 1) can be configured to use a single antenna and a single power amplifier and none of the transmitter devices need make use of active beam-forming control to wirelessly deliver sufficient power to the harvesters.

In some embodiments, the transmitter controller IC **1060** provides the viable RF power level directly to the one or more power amplifier units **1120** and does not use a beam-forming IC. In some embodiments, by not using beam-forming control, there is no active beam-forming control in the power transmission system. For example, in some

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embodiments, by eliminating the active beam-forming control, the relative phases of the power signals from different antennas are unaltered after transmission. In some embodiments, by eliminating the active beam-forming control, the phases of the power signals are not controlled and remain in a fixed or initial phase.

In some embodiments, respective antenna arrays **1010** of the one or more transmitters **1002** may include a set of one or more antennas configured to transmit the power waves **1016** into respective transmission fields of the one or more transmitters **1002**. Integrated circuits of the respective transmitter **1002**, such as a controller circuit (e.g., a radio frequency integrated circuit (RFIC)) and/or waveform generator, may control the behavior of the antennas. For example, based on the information received from the receiver **1020** by way of the communication signal **1018**, a controller circuit (e.g., processor **1004** of the transmitter **1002**, FIG. 12) may determine values of the waveform characteristics (e.g., amplitude, frequency, trajectory, direction, phase, polarization, among other characteristics) of power waves **1016** that would effectively provide power to the receiver **1020**, and in turn, the electronic device **1022**. The controller circuit may also identify a subset of antennas from the antenna arrays **1010** that would be effective in transmitting the power waves **1016**. In some embodiments, a waveform generator circuit (not shown in FIG. 10) of the respective transmitter **1002** coupled to the processor **1004** may convert energy and generate the power waves **1016** having the specific values for the waveform characteristics identified by the processor **1004**/controller circuit, and then provide the power waves to the antenna arrays **1010** for transmission.

In some embodiments, the communications component **1012** transmits data-communication signals **1018** by way of a wired and/or wireless communication connection to the receiver **1020**. In some embodiments, the communications component **1012** generates data-communication signals **1018** used for triangulation of the receiver **1020** (e.g., test signals). In some embodiments, the data-communication signals **1018** are used to convey information between the transmitter **1002** and receiver **1020** for adjusting values of one or more waveform characteristics used to transmit the power waves **1016** (e.g., convey amounts of power derived from RF test signals). In some embodiments, the data-communication signals **1018** include information related to status, efficiency, user data, power consumption, billing, geo-location, and other types of information.

In some embodiments, the communications component **1012** transmits data-communication signals **1018** to the receiver **1020** by way of the electronic device **1022a**. For example, the communications component **1012** may convey information to the communications component **1036** of the electronic device **1022a**, which the electronic device **1022a** may in turn convey to the receiver **1020** (e.g., via bus **1038**).

In some embodiments, the communications component **1012** includes a communications component antenna for communicating with the receiver **1020** and/or other transmitters **1002** (e.g., transmitters **1002b** through **1002n**). In some embodiments, these data-communication signals **1018** are sent using a first channel (e.g., a first frequency band) that is independent and distinct from a second channel (e.g., a second frequency band distinct from the first frequency band) used for transmission of the power waves **1016**.

In some embodiments, the receiver **1020** includes a receiver-side communications component **1044** configured to communicate various types of data with one or more of the transmitters **1002**, through a respective communication

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signal **1018** generated by the receiver-side communications component (in some embodiments, a respective communication signal **1018** is referred to as an advertising signal). The data may include location indicators for the receiver **1020** and/or electronic device **1022**, a power status of the device **1022**, status information for the receiver **1020**, status information for the electronic device **1022**, status information about the power waves **1016**, and/or status information for pockets of energy. In other words, the receiver **1020** may provide data to the transmitter **1002**, by way of the communication signal **1018**, regarding the current operation of the system **1000**, including: information identifying a present location of the receiver **1020** or the device **1022**, an amount of energy (i.e., usable power) received by the receiver **1020**, and an amount of power received and/or used by the electronic device **1022**, among other possible data points containing other types of information.

In some embodiments, the data contained within data-communication signals **1018** is used by the electronic device **1022**, the receiver **1020**, and/or the transmitters **1002** for determining adjustments to values of one or more waveform characteristics used by the antenna array **1010** to transmit the power waves **1016**. Using a communication signal **1018**, the transmitter **1002** communicates data that is used, e.g., to identify receivers **1020** within a transmission field, identify electronic devices **1022**, determine safe and effective waveform characteristics for power waves, and/or hone the placement of pockets of energy. In some embodiments, the receiver **1020** uses a communication signal **1018** to communicate data for, e.g., alerting transmitters **1002** that the receiver **1020** has entered or is about to enter a transmission field (e.g., come within wireless-power-transmission range of a transmitter **1002**), provide information about the electronic device **1022**, provide user information that corresponds to the electronic device **1022**, indicate the effectiveness of received power waves **1016**, and/or provide updated characteristics or transmission parameters that the one or more transmitters **1002** use to adjust transmission of the power waves **1016**.

In some embodiments, the receiver **1020** does not include a distinct communications component **1044**. Rather, the receiver **1020** is configured to reflect RF signals transmitted by the transmitter **1002** at the one or more antennas **1024** and, importantly, modulate the reflected RF signals to convey data (or a message) to transmitter **1002**. In such embodiments, the transmitter **1002** may also lack a distinct communications component. Instead, the transmitter **1002** may receive the reflected RF signals at the one or more antenna arrays **1010**, and the transmitter **1002** may demodulate the reflected RF signals in order to interpret them.

In some embodiments, transmitter sensor **1014** and/or receiver sensor **1028** detect and/or identify conditions of the electronic device **1022**, the receiver **1020**, the transmitter **1002**, and/or a transmission field. In some embodiments, data generated by the transmitter sensor **1014** and/or receiver sensor **1028** is used by the transmitter **1002** to determine appropriate adjustments to values of one or more waveform characteristics used to transmit the power waves **1016**. Data from transmitter sensor **1014** and/or receiver sensor **1028** received by the transmitter **1002** includes, e.g., raw sensor data and/or sensor data processed by a processor **1004**, such as a sensor processor. Processed sensor data includes, e.g., determinations based upon sensor data output. In some embodiments, sensor data received from sensors that are external to the receiver **1020** and the transmitters **1002** is also used (such as thermal imaging data, information from optical sensors, and others).

FIG. 11A is a block diagram of a wireless-power transmitter, in accordance with some embodiments. The block diagram of a wireless-power transmitter 1100 corresponds to an example of the components that can be included within the wireless-power transmitter 1002 described above in reference to FIG. 10. The wireless-power transmitter 1002 can be referred to herein as a near-field (NF) power transmitter device, transmitter, power transmitter, or wireless-power transmitter device. The wireless-power transmitter 1002 includes one or more of one or more communications components 1110, one or more power amplifier units 1120-1, . . . 1120-*n*, one or more power-transfer elements (e.g., such as antennas 1130-1 to 1130-*n* (which can be instances of the transmitter antenna elements 1036; FIGS. 1-12D)), an RF Power Transmitter Integrated Circuit (RFIC) 1160 (e.g., analogous to controller 1040 FIG. 10), and one or more sensors 1165.

In some embodiments, the communication component(s) 1110 (e.g., wireless communication components, such as WI-FI or BLUETOOTH radios) enable communication between the wireless-power transmitter 1100 and one or more communication networks. In some embodiments, the communication component(s) 1110 are capable of data communications using any of a variety of custom or standard wireless protocols (e.g., IEEE 802.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, MiWi, etc.) custom or standard wired protocols (e.g., Ethernet, HomePlug, etc.), and/or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

In some embodiments, the communication component(s) 1110 receives charging information from a wireless-power receiver (or from an electronic device configured to be charged by the wireless-power receiver; e.g., a wireless-power receiver 1055 described in reference to FIG. 13). In some embodiments, the charging information is received in a packet of information that is received in conjunction with an indication that the wireless-power receiver is located within one meter of the wireless-power transmitter 1035 in FIG. 14. In some embodiments, the charging information includes the location of the wireless-power receiver 1055 within the transmission field of the wireless-power transmitter 1035 (or the surrounding area within the communications component(s) range). For example, communication components 1110, such as BLE communications paths operating at 2.4 GHz, to enable the wireless-power transmitter 1035 to monitor and track the location of the wireless-power receiver 1055. The location of the wireless-power receiver 1055 can be monitored and tracked based on the charging information received from the wireless-power receiver 1055 via the communications components 1110.

In some embodiments, the charging information indicates that a wireless-power receiver 1055 is authorized to receive wirelessly-delivered power from the wireless-power transmitter 1035. More specifically, the wireless-power receiver can use a wireless communication protocol (such as BLE) to transmit the charging information as well as authentication information to the one or more integrated circuits (e.g., RFIC 1160) of the wireless-power transmitter 1035. In some embodiments, the charging information also includes general information such as charge requests from the receiver, the current battery level, charging rate (e.g., effectively transmitted power or electromagnetic energy successfully converted to usable energy), device specific information

(e.g., temperature, sensor data, receiver requirements or specifications, and/or other receiver specific information), etc.

In some instances, the communication component(s) 1110 are not able to communicate with wireless-power receivers for various reasons, e.g., because there is no power available for the communication component(s) 1110 to use for the transmission of data signals or because the wireless-power receiver itself does not actually include any communication component of its own. As such, in some embodiments, the wireless-power transmitters 1035 described herein are still able to uniquely identify different types of devices and, when a wireless-power receiver 1055 is detected, figure out if that the wireless-power receiver 1055 is authorized to receive wireless-power (e.g., by measuring impedances, reflected power, and/or other techniques).

The one or more power amplifiers 1120 are configured to amplify an electromagnetic signal that is provided to the one or more antennas 1130. In some embodiments, the power amplifier 1120 used in the power transmission system controls both the efficiency and gains of the output of the power amplifier. In some embodiments, the power amplifier used in the power transmission system is a class E power amplifier 1120. In some embodiments, the power amplifier 1120 used in the power transmission system is a Gallium Nitride (GaN) power amplifier. In some embodiments, the wireless-power transmitters 1035 is configured to control operation of the one or more power amplifiers 1120 when they drive one or more antennas 1130. In some embodiments, one or more of the power amplifiers 1120 are a variable power amplifier including at least two power levels. In some embodiments, a variable power amplifier includes one or more of a low power level, median power level, and high power level. As discussed below in further detail, in some embodiments, the wireless-power transmitters 1035 is configured to select power levels of the one or more power amplifiers. In some embodiments, the power (e.g., electromagnetic power) is controlled and modulated at the wireless-power transmitters 1035 via switch circuitry as to enable the wireless-power transmitters 1035 to send electromagnetic energy to one or more wireless receiving devices (e.g., wireless-power receivers 1055) via the one or more antennas 1130.

In some embodiments, the output power of the single power amplifier 1120 is equal or greater than 2 W. In some embodiments, the output power of the single power amplifier 1120 is equal or less than 15 W. In some embodiments, the output power of the single power amplifier 1120 is greater than 2 W and less than 15 W. In some embodiments, the output power of the single power amplifier 1120 is equal or greater than 4 W. In some embodiments, the output power of the single power amplifier 1120 is equal or less than 8 W. In some embodiments, the output power of the single power amplifier 1120 is greater than 4 W and less than 8 W. In some embodiments, the output power of the single power amplifier 1120 is greater than 8 W and up to 50 W.

In some embodiments, by using the single power amplifier 1120 with an output power range from 2 W to 15 W, the electric field within the power transmission range of the antenna 1130 controlled by the single power amplifier 1120 is at or below a SAR value of 1.6 W/kg, which is in compliance with the FCC (Federal Communications Commission) SAR requirement in the United States. In some embodiments, by using a single power amplifier 1120 with a power range from 2 W to 15 W, the electric field within the power transmission range of the antenna 1130 controlled by the single power amplifier 1120 is at or below a SAR value of 2 W/kg, which is in compliance with the IEC (Internation-

tional Electrotechnical Commission) SAR requirement in the European Union. In some embodiments, by using a single power amplifier **1120** with a power range from 2 W to 15 W, the electric field within the power transmission range of the antenna **1130** controlled by the single power amplifier **1120** is at or below a SAR value of 0.8 W/kg. In some embodiments, by using a single power amplifier **1120** with a power range from 2 W to 15 W, the electric field within the power transmission range of the antenna **1130** controlled by the single power amplifier **1120** is at or below any level that is regulated by relevant rules or regulations. In some embodiments, the SAR value in a location of the radiation profile of the antenna decreases as the range of the radiation profile increases.

In some embodiments, the radiation profile generated by the antenna controlled by a single power amplifier is defined based on how much usable power is available to a wireless-power receiver when it receives electromagnetic energy from the radiation profile (e.g., rectifies and converts the electromagnetic energy into a usable DC current), and the amount of usable power available to such a wireless-power receiver **1055** can be referred to as the effective transmitted power of an electromagnetic signal. In some embodiments, the effective transmitted power of the electromagnetic signal in a predefined radiation profile is at least 0.5 W. In some embodiments, the effective transmitted power of the signal in a predefined radiation profile is greater than 1 W. In some embodiments, the effective transmitted power of the signal in a predefined radiation profile is greater than 2 W. In some embodiments, the effective transmitted power of the signal in a predefined radiation profile is greater than 5 W. In some embodiments, the effective transmitted power of the signal in a predefined radiation profile is less or equal to 4 W.

FIG. 11B is a block diagram of another wireless-power transmitter **1150** (e.g., wireless-power receiver **1035**) including an RF power transmitter integrated circuit **1160**, one or more sensors **1165**, one or more antennas **1130**, and/or a power amplifier **1120** in accordance with some embodiments. For case of discussion and illustration, the other wireless-power transmitters **1150** can be an instance of the wireless-power transmitter devices described above in reference to FIGS. 1-13, and includes one or more additional and/or distinct components, or omits one or more components. In some embodiments, the RFIC **1160** includes a CPU subsystem **1170**, an external device control interface, a subsection for DC to power conversion, and analog and digital control interfaces interconnected via an interconnection component, such as a bus or interconnection fabric block **1171**. In some embodiments, the CPU subsystem **1170** includes a microprocessor unit (CPU) **1173** with related Read-Only-Memory (ROM) **1172** for device program booting via a digital control interface, e.g., an I2C port, to an external FLASH containing the CPU executable code to be loaded into the CPU Subsystem Random Access Memory (RAM) **1174** (e.g., memory **1006**, FIG. 10) or executed directly from FLASH. In some embodiments, the CPU subsystem **1370** also includes an encryption module or block **1176** to authenticate and secure communication exchanges with external devices, such as wireless-power receivers that attempt to receive wirelessly delivered power from the Wireless-power transmitters **1035**. In some embodiments, the wireless-power transmitters **1035** may also include a temperature monitoring circuit (not shown) that is in communication with the CPU subsystem **1170** to ensure that the wireless-power transmitters **1035** remains within an acceptable temperature range. For example, if a determination is made that the wireless-power transmitters **1035** has reached

a threshold temperature, then operation of the wireless-power transmitters **1035** may be temporarily suspended until the wireless-power transmitters **1035** falls below the threshold temperature.

In some embodiments, the RFIC **1160** also includes (or is in communication with) a power amplifier controller IC (PAIC) **1161A** that is responsible for controlling and managing operations of a power amplifier, including, but not limited to, reading measurements of impedance at various measurement points within the power amplifier, instructing the power amplifier to amplify the electromagnetic signal, synchronizing the turn on and/or shutdown of the power amplifier, optimizing performance of the power amplifier, protecting the power amplifier, and other functions discussed herein. In some embodiments, the impedance measurement are used to allow the wireless-power transmitters **1035** (via the RFIC **1160** and/or PAIC **1161A**) to detect of one or more foreign objects, optimize operation of the one or more power amplifiers, assess one or more safety thresholds, detect changes in the impedance at the one or more power amplifiers, detect movement of the receiver within the wireless transmission field, protect the power amplifier from damage (e.g., by shutting down the power amplifier, changing a selected power level of the power amplifier, and/or changing other configurations of the wireless-power transmitters **1035**), classify a receiver (e.g., authorized receivers, unauthorized receivers, and/or receiver with an object), compensate for the power amplifier (e.g., by making hardware, software, and/or firmware adjustments), tune the wireless-power transmitters **1035**, and/or other functions.

In some embodiments, the PAIC **1161A** may be on the same integrated circuit as the RFIC **1160**. Alternatively, in some embodiments, the PAIC **1161A** may be on its own integrated circuit that is separate from (but still in communication with) the RFIC **1160**. In some embodiments, the PAIC **1161A** is on the same chip with one or more of the power amplifiers **1120**. In some other embodiments, the PAIC **1161A** is on its own chip that is a separate chip from the power amplifiers **1320**. In some embodiments, the PAIC **1161A** may be on its own integrated circuit that is separate from (but still in communication with) the RFIC **1160** enables older systems to be retrofitted. In some embodiments, the PAIC **1161A** as a standalone chip communicatively coupled to the RFIC **1160** can reduce the processing load and potential damage from over-heating. Alternatively or additionally, in some embodiments, it is more efficient to design and use two different ICs (e.g., the RFIC **1160** and the PAIC **1161A**).

In some embodiments, executable instructions running on the CPU (such as those shown in the memory **1206** in FIG. 12, and described below) are used to manage operation of the wireless-power transmitters **1035** and to control external devices through a control interface, e.g., SPI control interface **1175**, and the other analog and digital interfaces included in the RFIC **1160**. In some embodiments, the CPU subsystem **1170** also manages operation of the subsection of the RFIC **1160**, which includes a local oscillator (LO) **1177** and a transmitter (TX) **1178**. In some embodiments, the LO **1177** is adjusted based on instructions from the CPU subsystem **1170** and is thereby set to different desired frequencies of operation, while the TX converts, amplifies, modulates the output as desired to generate a viable power level.

In some embodiments, the RFIC **1360** and/or PAIC **1161A** provide the viable power level (e.g., via the TX **1178**) directly to the one or more power amplifiers **1120** and does not use any beam-forming capabilities (e.g., bypasses/disables a beam-forming IC and/or any associated algorithms if

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phase-shifting is not required, such as when only a single antenna **1130** is used to transmit power transmission signals to a wireless-power receiver **1055**). In some embodiments, by not using beam-forming control, there is no active beam-forming control in the power transmission system. For example, in some embodiments, by eliminating the active beam-forming control, the relative phases of the power signals from different antennas are unaltered after transmission. In some embodiments, by eliminating the active beam-forming control, the phases of the power signals are not controlled and remain in a fixed or initial phase. In some embodiments, the RFIC **1160** and/or PAIC **1161A** regulate the functionality of the power amplifiers **1120** including adjusting the viable power level to the power amplifiers **1120**, enabling the power amplifiers **1120**, disabling the power amplifiers **1120**, and/or other functions.

Various arrangements and couplings of power amplifiers **1120** to antenna coverage areas **1190** allow the wireless-power receiver **1055** to sequentially or selectively activate different antenna coverage areas **1190** (i.e., power transfer points) in order to determine the most efficient and safest (if any) antenna coverage area **1190** to use for transmitting wireless-power to a wireless-power receiver **1055**.

In some embodiments, the one or more power amplifiers **1120** are also controlled by the CPU subsystem **1170** to allow the CPU **1173** to measure output power provided by the power amplifiers **1120** to the antenna coverage areas (i.e., plurality of power-transfer points **202**) of the wireless-power transmitter **1035**. In some embodiments, the one or more power amplifiers **1120** are controlled by the CPU subsystem **1170** via the PAIC **1161A**. In some embodiments, the power amplifiers **1120** may include various measurement points that allow for at least measuring impedance values that are used to enable the foreign object detection techniques, receiver and/or foreign object movement detection techniques, power amplifier optimization techniques, power amplifier protection techniques, receiver classification techniques, power amplifier impedance detection techniques, and/or other safety techniques described in commonly-owned U.S. patent application Ser. No. 16/932,631, which is incorporated by reference in its entirety for all purposes.

FIG. **12** is a block diagram illustrating one or more components of a wireless-power transmitter **1035**, in accordance with some embodiments. In some embodiments, the wireless-power transmitter **1035** includes an RFIC **1160** (and the components included therein, such as a PAIC **1161A** and others described above in reference to FIGS. **11A-11B**), memory **1206** (which may be included as part of the RFIC **1160**, such as nonvolatile memory **1206** that is part of the CPU subsystem **1170**), one or more CPUs **1173**, and one or more communication buses **1208** for interconnecting these components (sometimes called a chipset). In some embodiments, the wireless-power transmitter **1035** includes one or more sensors **1165**. In some embodiments, the wireless-power transmitter **1035** includes one or more output devices such as one or more indicator lights, a sound card, a speaker, a small display for displaying textual information and error codes, etc. In some embodiments, the wireless-power transmitter **1035** includes a location detection device, such as a GPS other geo-location receiver, for determining the location of the wireless-power transmitter **1035**.

In some embodiments, the one or more sensors **1165** include one or more capacitive sensors, inductive sensors, ultrasound sensors, photoelectric sensors, time-of-flight sensors (e.g., IR sensors, ultrasonic time-of-flight sensors, phototransistor receiver systems, etc.), thermal radiation sen-

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sors, ambient temperature sensors, humidity sensors, IR sensors or IR LED emitter, occupancy sensors (e.g., RFID sensors), ambient light sensors, motion detectors, accelerometers, heat detectors, hall sensors, proximity sensors, sound sensors, pressure detectors, light and/or image sensors, and/or gyroscopes, as well as integrated sensors in one or more antennas.

In some embodiments, the wireless-power transmitter **1035** further includes an optional signature-signal receiving circuit **1240**, an optional reflected power coupler **1248**, and an optional capacitive charging coupler **1250**.

The memory **1406** includes high-speed random access memory, such as DRAM, SRAM, DDR SRAM, or other random access solid state memory devices; and, optionally, includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. The memory **1206**, or alternatively the non-volatile memory within memory **1206**, includes a non-transitory computer-readable storage medium. In some embodiments, the memory **1206**, or the non-transitory computer-readable storage medium of the memory **1406**, stores the following programs, modules, and data structures, or a subset or superset thereof:

- Operating logic **1216** including procedures for handling various basic system services and for performing hardware dependent tasks;

- Communication module **1228** for coupling to and/or communicating with remote devices (e.g., remote sensors, transmitters, receivers, servers, mapping memories, etc.) in conjunction with wireless communication component(s) **1110**;

- Sensor module **1220** for obtaining and processing sensor data (e.g., in conjunction with sensor(s) **1365**) to, for example, determine or detect the presence, velocity, and/or positioning of object in the vicinity of the wireless-power transmitter **1035** as well as classify a detected object;

- Power-wave generating module **1222** for generating and transmitting power transmission signals (e.g., in conjunction with antenna coverage areas **1190** and the antennas **1130** respectively included therein), including but not limited to, forming pocket(s) of energy at given locations, and controlling and/or managing the power amplifier (e.g., by performing one or functions of the PAIC **1361A**). Optionally, the power-wave generating module **1222** may also be used to modify values of transmission characteristics (e.g., power level (i.e., amplitude), phase, frequency, etc.) used to transmit power transmission signals by individual antenna coverage areas;

- Impedance determining module **1223** for determining an impedance of the power amplifier based on parametric parameters obtained from one or more measurement points within the wireless-power transmitter **1035** (e.g., determining an impedance using one or more Smith charts). Impedance determining module **1223** may also be used to determine the presence of a foreign object, classify a receiver, detect changes in impedances, detect movement of a foreign object and/or receiver, determine optimal and/or operational impedances, as well as a number of other functions describe below;

- Database **1224**, including but not limited to:

- Sensor information **1226** for storing and managing data received, detected, and/or transmitted by one or more sensors (e.g., sensors **1165** and/or one or more remote sensors);

Device settings **1228** for storing operational settings for the wireless-power transmitter **1035** and/or one or more remote devices including, but not limited to, lookup tables (LUT)s for SAR, e-field roll-off, producing a certain radiation profile from among various radiation profiles, Smith Charts, antenna tuning parameters, and/or values associated with parametric parameters of the wireless-power transmitter **1035** for different configurations (e.g., obtained during simulation, characterization, and/or manufacture tests of the wireless-power transmitter **1035** and/or updated during operation (e.g., learned improvements to the system)). Alternatively, raw values can be stored for future analysis;

Communication protocol information **1230** for storing and managing protocol information for one or more protocols (e.g., custom or standard wireless protocols, such as ZigBee, Z-Wave, etc. and/or custom or standard wired protocols, such as Ethernet); and

Optional learned signature signals **1232** for a variety of different wireless-power receivers and other objects (which are not wireless-power receivers).

A secure element module **1234** for determining whether a wireless-power receiver is authorized to receive wirelessly delivered power from the wireless-power transmitter **1035**;

An antenna zone selection and tuning module **1237** for coordinating a process of transmitting test power transmission signals to an antenna **1230** (e.g., antenna element **1036**) with various antenna coverage areas (i.e., power-transfer points) to determine which antenna coverage area (i.e., power-transfer point) should be used to wirelessly deliver power to various wireless-power receivers as described herein (additional examples and embodiments are provided in reference to FIGS. 9A-9B of PCT Patent Application No. PCT/US2019/015820 (U.S. Pat. No. 10,615,647), which is incorporated by reference in its entirety for all purposes; and also provided in PCT/US2017/065886 (U.S. Pat. No. 10,256,677), which is incorporated by reference in its entirety for all purposes);

An authorized receiver and object detection module **1238** used for detecting various signature signals from wireless-power receivers and from other objects, and then determining appropriate actions based on the detecting of the various signature signals (as is explained in more detail in reference to FIGS. 9A-9B of PCT Patent Application No. PCT/US2019/015820 (U.S. Pat. No. 10,615,647), which is incorporated by reference in its entirety for all purposes; also explained in more detail in PCT/US2017/065886 (U.S. Pat. No. 10,256,677), which is incorporated by reference in its entirety for all purposes); and

An optional signature-signal decoding module **1239** used to decode the detected signature signals and determine message or data content. In some embodiments, the module **1239** includes an electrical measurement module **1242** to collect electrical measurements from one or more receivers (e.g., in response to power beacon signals), a feature vector module **1244** to compute feature vectors based on the electrical measurements collected by the electrical measurement module **1239**, and/or machine learning classifier model(s) **1246** that are trained to detect and/or classify foreign objects (additional detail provided in commonly-owned U.S. Patent Publication No. 2019/0245389, which is incorporated by reference herein for all purposes).

Each of the above-identified elements (e.g., modules stored in memory **1206** of the wireless-power transmitter **1035**) is optionally stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing the function(s) described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are optionally combined or otherwise rearranged in various embodiments. In some embodiments, the memory **1206**, optionally, stores a subset of the modules and data structures identified above.

FIG. 13 is a block diagram illustrating a representative wireless-power receiver **1055** (also sometimes interchangeably referred to herein as a receiver, or power receiver), in accordance with some embodiments. In some embodiments, the wireless-power receiver **1055** includes one or more processing units (e.g., CPUs, ASICs, FPGAs, microprocessors, and the like) **1352**, one or more communication components **1354**, memory **1356**, antenna(s) **1360** (which can be instances receiver antenna elements **104**; FIGS. 1A-3B), power harvesting circuitry **1359** (e.g., PCB **108**; FIGS. 1A-3B), and one or more communication buses **1358** for interconnecting these components (sometimes called a chip-set). In some embodiments, the wireless-power receiver **1055** includes one or more optional sensors **1362**, similar to the one or sensors **1028** and **1165** described above with reference to FIGS. 10 and 11, respectively. In some embodiments, the wireless-power receiver **1055** includes an energy storage device **1361** for storing energy harvested via the power harvesting circuitry **1359**. In various embodiments, the energy storage device **1361** includes one or more batteries, one or more capacitors, one or more inductors, and the like.

In some embodiments, the power harvesting circuitry **1359** includes one or more rectifying circuits and/or one or more power converters. In some embodiments, the power harvesting circuitry **1359** includes one or more components (e.g., a power converter) configured to convert energy from power waves and/or energy pockets to electrical energy (e.g., electricity). In some embodiments, the power harvesting circuitry **1359** is further configured to supply power to a coupled electronic device, such as a laptop or phone. In some embodiments, supplying power to a coupled electronic device include translating electrical energy from an AC form to a DC form (e.g., usable by the electronic device).

In some embodiments, the optional signature-signal generating circuit **1310** includes one or more components as discussed with reference to FIGS. 3A-3D of commonly-owned U.S. Patent Publication No. 2019/0245389, which is incorporated by reference in its entirety for all purposes.

In some embodiments, the antenna(s) **1360** include one or more helical antennas, such as those described in detail in commonly-owned U.S. Pat. No. 10,734,717, which is incorporated by reference in its entirety for all purposes (e.g., with particular reference to FIGS. 2-4B, and elsewhere).

In some embodiments, the wireless-power receiver **1055** includes one or more output devices such as one or more indicator lights, a sound card, a speaker, a small display for displaying textual information and error codes, etc. In some embodiments, the wireless-power receiver **1055** includes a location detection device, such as a GPS (global positioning satellite) or other geo-location receiver, for determining the location of the wireless-power transmitter **1055**.

In various embodiments, the one or more sensors **1362** include one or more thermal radiation sensors, ambient temperature sensors, humidity sensors, IR sensors, occu-

pancy sensors (e.g., RFID sensors), ambient light sensors, motion detectors, accelerometers, and/or gyroscopes. It is noted that the foreign object detection techniques can operate without relying on the one or more sensor(s) **1362**.

The communication component(s) **1354** enable communication between the wireless-power receiver **1055** and one or more communication networks. In some embodiments, the communication component(s) **1354** are capable of data communications using any of a variety of custom or standard wireless protocols (e.g., IEEE 802.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, MiWi, etc.) custom or standard wired protocols (e.g., Ethernet, HomePlug, etc.), and/or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document. It is noted that the foreign object detection techniques can operate without relying on the communication component(s) **1354**.

The communication component(s) **1354** include, for example, hardware capable of data communications using any of a variety of custom or standard wireless protocols (e.g., IEEE 802.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, MiWi, etc.) and/or any of a variety of custom or standard wired protocols (e.g., Ethernet, HomePlug, etc.), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

The memory **1356** includes high-speed random access memory, such as DRAM, SRAM, DDR SRAM, or other random access solid state memory devices; and, optionally, includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. The memory **1356**, or alternatively the non-volatile memory within memory **1356**, includes a non-transitory computer-readable storage medium. In some embodiments, the memory **1356**, or the non-transitory computer-readable storage medium of the memory **1356**, stores the following programs, modules, and data structures, or a subset or superset thereof:

Operating logic **1366** including procedures for handling various basic system services and for performing hardware dependent tasks;

Communication module **1368** for coupling to and/or communicating with remote devices (e.g., remote sensors, transmitters, receivers, servers, mapping memories, etc.) in conjunction with communication component(s) **1354**;

Optional sensor module **1370** for obtaining and processing sensor data (e.g., in conjunction with sensor(s) **1362**) to, for example, determine the presence, velocity, and/or positioning of the wireless-power receiver **1055**, a wireless-power transmitter **1055**, or an object in the vicinity of the wireless-power transmitter **1055**;

Wireless power-receiving module **1372** for receiving (e.g., in conjunction with antenna(s) **1360** and/or power harvesting circuitry **1359**) energy from, capacitively-conveyed electrical signals, power waves, and/or energy pockets; optionally converting (e.g., in conjunction with power harvesting circuitry **1359**) the energy (e.g., to direct current); transferring the energy to a coupled electronic device; and optionally storing the energy (e.g., in conjunction with energy storage device **1361**);

Database **1374**, including but not limited to:

Sensor information **1376** for storing and managing data received, detected, and/or transmitted by one or more sensors (e.g., sensors **1362** and/or one or more remote sensors);

Device settings **1378** for storing operational settings for the wireless-power transmitter **1055**, a coupled electronic device, and/or one or more remote devices; and

Communication protocol information **1380** for storing and managing protocol information for one or more protocols (e.g., custom or standard wireless protocols, such as ZigBee, Z-Wave, etc. and/or custom or standard wired protocols, such as Ethernet);

A secure element module **1382** for providing identification information to the wireless-power transmitter **1035** (e.g., the wireless-power transmitter **1035** uses the identification information to determine if the wireless-power receiver **1304** is authorized to receive wirelessly delivered power); and

An optional signature-signal generating module **1383** used to control (in conjunction with the signature-signal generating circuit **1310**) various components to cause impedance changes at the antenna(s) **1360** and/or power harvesting circuitry **1359** to then cause changes in reflected power as received by a signature-signal receiving circuit **1240**.

Each of the above-identified elements (e.g., modules stored in memory **1356** of the receiver **1304**) is optionally stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing the function(s) described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are optionally combined or otherwise rearranged in various embodiments. In some embodiments, the memory **1356**, optionally, stores a subset of the modules and data structures identified above. Furthermore, the memory **1356**, optionally, stores additional modules and data structures not described above, such as an identifying module for identifying a device type of a connected device (e.g., a device type for an electronic device that is coupled with the receiver **1304**).

In some embodiments, the near-field power transmitters disclosed herein may use adaptive loading techniques to optimize power transfer. Such techniques are described in detail in commonly-owned and incorporated-by-reference PCT Application No. PCT/US2017/065886 and, in particular, in reference to FIGS. 5-8 and 12-15 of PCT Application No. PCT/US2017/065886.

In some embodiments, the wireless-power transmitter **1055** is coupled to or integrated with an electronic device, such as shelving unit, a pen, a marker, a phone, a tablet, a laptop, a hearing aid, smart glasses, headphones, computer accessories (e.g., mouse, keyboard, remote speakers), and/or other electrical devices. In some embodiments, the wireless-power transmitter **1055** is coupled to or integrated with small consumer device, such as a fitness band, a smart watch, and/or other wearable product. Alternatively, in some embodiments, the wireless-power transmitter **1055** is an electronic device.

In accordance with some embodiments, a wireless-power harvester integrated in a small form-factor device (e.g., less than or equal to 100 mm in length, 70 mm in width and 30 mm in depth) (e.g., 50-100 mm in length, 35-70 mm in width, and 15-30 mm in depth) (e.g., an example small form-factor device such as the electronic device **102** shown in FIGS. 1A-1B, 2, and 3A-3B), comprises a stamped metal

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harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape (e.g., FIGS. 1A-1B, 2, and 3A-3B each illustrate harvester antenna **104** that follows a meandering path). The meandering shape includes a predetermined number of turns (e.g., FIGS. 1A-1B, 2, and 3A-3B each illustrate harvester antenna **104** having at least two turns). Specifically, FIG. 1B shows in a bounding box a first turn **161**. A first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB configured to operate as a ground plane for the stamped metal antenna (e.g., FIGS. 1B and 2 illustrate harvester antenna **104** coupled to the PCB **108** of the electronic device **102** and a receiver integrated circuit **156**). An intermediate portion, disposed between the first end and the second end, of the meandering shape is coupled to power-conversion circuitry that is separate from the PCB, the power-conversion circuitry is configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device (e.g., FIGS. 1B and 2 illustrate an intermediate portion **164** of the receiving antenna **104** being coupled to a receiver integrated circuit **156**, and FIG. 10 illustrates a battery **1030**).

In some embodiments, the small form-factor device is a digital price tag including a display (e.g., an e-ink display, an LCD display, etc.) powered by the battery. The display and the battery being coupled to the PCB. For example, FIG. 1A illustrates that the electronic device **102** includes a display **106** that is configured to be a digital price tag.

In some embodiments, the small form-factor device is a digital thermometer powered by the battery. In some embodiments, the thermometer can include a display and/or provide data to a computer or server. In some embodiments, other measurement or stock tracking devices can be used. For example, a scale, a counter (e.g., identifying a number of object on a shelf). In some embodiments, shopping carts with displays can also be included.

In some embodiments, the power-conversion circuitry is on a substrate distinct from the PCB.

In some embodiments, the stamped metal harvesting antenna is quasi omnidirectional. For example, FIG. 1B illustrates that the harvester antenna **104** is quasi omnidirectional and configured to receive wireless power with any polarization. In some embodiments, the antenna is configured to receive wireless power with any polarization.

In some embodiments, the stamped metal harvesting antenna is coupled to an edge of the PCB. For example, FIGS. 1B and 2 illustrate harvester antenna **104** coupled to the PCB **108** of the electronic device **102** and a receiver integrated circuit **156**. In some embodiments, the harvesting antenna can be attached to any edge portion of the PCB.

In some embodiments, the stamped metal harvesting antenna is modular, such that it can be interchangeably coupled with the PCB and the power-conversion circuitry at a first point in time, and separately coupled with a different PCB of a different small form-factor device and different power-conversion circuitry at a second point in time that is distinct from the first point in time. In some embodiments, the harvesting antenna can be designed to accommodate any configuration of the small form-factor device. In some embodiments, the harvesting antenna is easy and inexpensive to manufacture. In some embodiments, the harvesting

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antenna can be installed by either the manufacturer of the harvesting antenna or the purchaser of the harvesting antenna (e.g., the client).

In some embodiments, the stamped metal antenna occupies a first area of the housing of the small form-factor device that is adjacent to a second area of the housing occupied by the PCB, and the first and second areas are non-overlapping. For example, FIGS. 1B, 2, and 3A-3B illustrates the PCB **108** being in a non-overlapping adjacent location to the antenna **104**. Specifically, FIG. 1B shows a first area **109** that contains the PCB **108**, and a second area **111** that contains the antenna **104** and related componentry.

In some embodiments, the RF power waves have a centering operating frequency of 918 MHz.

In some embodiments, the stamped metal harvesting antenna has a gain of at least 2 dB. For example, FIG. 4 illustrates a harvesting antenna radiation pattern having a gain of at least 2 dB.

In some embodiments, the predetermined number of turns is two. For example, FIGS. 1A-1B, 2, and 3A-3B each illustrate harvester antenna **104** having at least two turns.

In some embodiments, the battery is a 60 to 100 mAh battery (e.g., battery **1030** in FIG. 10).

In some embodiments, the stamped metal harvesting antenna has a thickness of 1 mm to 2 mm.

In some embodiments, the stamped metal harvesting antenna has a width of 0.5 inch to 2 inches.

In some embodiments, the small form-factor device includes a communications component that is coupled to the PCB, the communication component configured to receive data that allows the small form-factor device to display graphical information. For example, electronic device **102** includes a communications component (e.g., communications component **1036** and/or **1044**; FIG. 10). In some embodiments, the data regarding the number of objects on a shelf is displayed. In some embodiments, the display can display information for updating a price or other information (e.g., product name or description). In some embodiments, the display can be used to display an estimate of the price of a virtual shopping cart/list of a shopper in front of the display.

In some embodiments, the graphical information is displayed using a text-only display of the small form-factor device. In some embodiments, the small form-factor device is further configured to communicatively couple to a wireless-power transmitter and provide charging information (e.g., battery life, amount of power received, charge requests, etc.).

In another aspect, a small form-factor device (e.g., less than or equal to 100 mm in length, 70 mm in width and 30 mm in depth) (e.g., 50-100 mm in length, 35-70 mm in width, and 15-30 mm in depth), comprises a wireless-power harvester. The wireless-power harvester includes a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape. The meandering shape includes a predetermined number of turns, a first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB is configured to operate as a ground plane for the stamped metal antenna. In some embodiments, an intermediate portion, disposed between the first end and the second end, of the meandering shape is coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry configured to

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convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device.

In another aspect, a shelving-mounted wireless-power-transmitting device is configured to transmit RF power waves to a wireless-power harvester device that is integrated in a small form-factor device. For example, FIG. 5 illustrates a front view 500 of a shelving system 501 that includes a shelving-mounted wireless-power transmitter system 504 placed at the top of the shelving system 501 for transmitting RF power waves to one or more electronic devices 102. In some embodiments, the wireless-power harvester device integrated in the small form-factor device (e.g., less than or equal to 100 mm in length, 70 mm in width and 30 mm in depth) (e.g., 50-100 mm in length, 35-70 mm in width, and 15-30 mm in depth) (e.g., an example small form-factor device such as the electronic device 102 shown in FIGS. 1A-1B, 2, and 3A-3B), includes a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape (e.g., FIGS. 1A-1B, 2, and 3A-3B each illustrate harvester antenna 104 that follows a meandering path). The meandering shape of the stamped metal harvesting antenna includes a predetermined number of turns. The stamped metal harvesting antenna includes a first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB configured to operate as a ground plane for the stamped metal antenna (e.g., FIGS. 1B and 2 illustrate harvester antenna 104 coupled to the PCB 108 of the electronic device 102 and a receiver integrated circuit 156). An intermediate portion, disposed between the first end and the second end, of the meandering shape is coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device (e.g., FIGS. 1B and 2 illustrate an intermediate portion 164 of the receiving antenna 104 being coupled to a receiver integrated circuit 156, and FIG. 10 illustrates a battery 1030).

In some embodiments, the shelving-mounted wireless-power-transmitting device comprises a mounting structure (e.g., mounting structure 503 shown at least in FIG. 5) that is mountable to a shelving unit (e.g., shelving system 501 shown in FIGS. 5-7). The shelving unit has a predetermined height and a predetermined width. The mounting includes a first end coupled (or integrated) to an outer portion of the shelving unit and a second end opposite the first end extending a predetermined distance away from the outer portion of the shelving unit. For example, FIGS. 5-7 illustrate shelving-mounted wireless power transmitting systems with shelving systems having predetermined height and widths. The second end includes a ground plane (dimensions of the ground plane are based on the size of the transmitting device; e.g., 8 inches by 2 inches). A first plurality of conductive segments has a first shape and a first size (e.g., FIG. 7 shows conductive segments of three different sizes (e.g., a first number of conductive segments 704, a second number of conductive segments 708, and third number of conductive segments 720)), each of the first plurality of conductive segments is disposed over the ground plane. A second plurality of conductive segments have a second

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shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size. A respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another (e.g., a first number of conductive segments 704, a second number of conductive segments 708, and third number of conductive segments 720). The first and second pluralities of conductive segments are disposed over the ground plane to form an antenna that is configured to radiate radio-frequency (RF) wireless power waves towards the shelving unit such that (i) the predetermined height and the predetermined width of the shelving unit are within a coverage area (e.g., up to 15 feet in a first direction (e.g., a horizontal direction and up to 20 feet in a second direction (e.g., a vertical direction along the height of the shelving unit) of the RF wireless power waves, and (ii) a bottom shelf of the shelving unit receives at least a threshold amount of RF energy from the RF wireless power waves. For example, at least enough RF energy to allow for a harvester to harvest enough usable power and/or charge for operating a digital price tag (e.g., an e-ink display and associated hardware). For example, FIG. 7 shows that the first coverage area 706, the second coverage area 714, and the third coverage area 722 each show the bottom shelf receiving at least a threshold amount of RF energy. In some embodiments, the usable power can be between 0.5 milliwatts and to over few tenths of Watt.

In accordance with some embodiments, a shelving-mounted wireless-power-transmitting device comprises a mounting structure (e.g., mounting structure 503 shown at least in FIG. 5) that is mountable to a shelving unit (e.g., shelving system 501 shown in FIGS. 5-7). The shelving unit has a predetermined height and a predetermined width. The mounting includes a first end coupled or integrated to an outer portion of the shelving unit and a second end opposite the first end extending a predetermined distance away from the outer portion of the shelving unit. For example, FIGS. 5-7 illustrate shelving-mounted wireless power transmitting systems with shelving systems having predetermined height and widths. The second end includes a ground plane (e.g., dimensions of the ground plane are based on the size of the transmitting device; e.g., 8 inches by 2 inches). The second end includes a first plurality of conductive segments having a first shape and a first size. For example, FIG. 7 shows conductive segments of three different sizes (e.g., a first number of conductive segments 704, a second number of conductive segments 708, and third number of conductive segments 720). The second end includes a second plurality of conductive segments that have a second shape and a second size, the second shape is distinct from the first shape and the second size is distinct from the first size. A respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another. For example, a first number of conductive segments 704, a second number of conductive segments 708, and third number of conductive segments 720. In some embodiments, the second end includes the first and second pluralities of conductive segments that are disposed over the ground plane to form an antenna that is configured to radiate radio-frequency (RF) wireless power waves towards the shelving unit such that (i) the predetermined height and the predetermined width of the shelving unit are within a coverage area (e.g., up to 15 feet in a first direction (e.g., a horizontal direction and up to 20 feet in a second direction (e.g., a vertical direction) of the RF wireless power waves, and (ii)

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a bottom shelf of the shelving unit receives at least a threshold amount of RF energy from the RF wireless power waves (e.g., at least enough RF energy to allow for a harvester to harvest enough usable power and/or charge for operating a digital price tag (e.g., an e-ink display and associated hardware)). For example, FIG. 7 shows that the first coverage area **706**, the second coverage area **714**, and the third coverage area **722** each show the bottom shelf receives at least a threshold amount of RF energy. In some embodiments, the usable power can be between 0.5 milli-watts and few tenths of a watt.

In some embodiments, each conductive segment of the first plurality of conductive segments and the second plurality of conductive segments are separated by a predetermined gap. In some embodiments, the gap is between 0.1 inches and 0.5 inches. The gap distance is determined based on the predetermined height and the predetermined width of the shelving structure. In some embodiments, each segment of the first plurality of conductive segments or each segment of the second plurality of conductive segments are coupled together via a feedline. In some embodiments, each conductive segment has a length and width. In some embodiments, the conductive segment length is approximately 1.5 inches and the antenna element width is approximately 0.5 inches. In some embodiments, the conductive segment length and width is determined based on the predetermined height and the predetermined width of the shelving unit's structure.

In some embodiments, the first shape and the second shape is a meandering path that produces a rectangular shape (e.g., the first shape is a meandering 'U' shape as shown in FIG. 7 as first number of conductive segments **704**, second number of conductive segments **708**, and third number of conductive segments **716**). In some embodiments, the conductive segments are a continuous piece of metal.

In some embodiments, the first size is larger than the second size. In some embodiments, the size is adjusted depending on the required coverage area (e.g., a larger shelf coverage area corresponds to a larger sized plurality of conductive segments).

In some embodiments, the first plurality conductive segments and the second plurality of conductive segments each include a predetermined number of two or more conductive segments (e.g., a first number of conductive segments **704**, a second number of conductive segments **708**, and third number of conductive segments **720**). In some embodiments, the predetermined number of conductive segments is between 5 to 15 antenna elements. In some embodiments, more than 15 antenna elements can be used. In some embodiments, the predetermined number of antenna elements is based on the predetermined height and the predetermined width of the shelving unit's structure.

In some embodiments, the outer portion of the shelving structure is a centrally located between two posts of the shelving structure. For example, FIG. 5 and FIG. 7 show the outer portion (e.g., mounting structure **503**) of the shelving structure being centrally located.

In some embodiments, respective numbers of segments in the first and second pluralities of conductive segments are adjustable to allow for creating an altered coverage area for the shelving-mounted wireless-power transmitting device.

In some embodiments, a peak gain of the antenna is at least 5.5 dBi and the wireless power waves are radiated using a center operating frequency of 918 MHz (e.g., FIG. 6 shows the peak gain being at least 5.5 dBi).

In some embodiments, the first plurality of conductive segments and the second plurality of conductive segments are formed using respective stamped copper plates.

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In some embodiments, the first plurality of conductive segments includes a first number of conductive segments, the first number of conductive segments selected based on the predetermined height and width of the shelving unit. For example, FIG. 7 shows conductive segments of three different sizes (e.g., a first number of conductive segments **704**, a second number of conductive segments **708**, and third number of conductive segments **720**). The second plurality of conductive segments includes a second number of conductive segments, the second number also selected based on the predetermined height and width of the shelving unit. For example, FIG. 7 shows conductive segments of three different sizes (e.g., a first number of conductive segments **704**, a second number of conductive segments **708**, and third number of conductive segments **720**).

In some embodiments, shelving-mounted wireless-power-transmitting device further comprises a second mounting structure that is mountable to a second shelving unit, the second shelving unit having a larger width and height than the predetermined height and width of the shelving unit, and the second mounting structure having a ground plane, a third plurality of conductive segments with each adjacent conductive segment of the third plurality is separated by a conductive segment of a fourth plurality of conductive segments. The third and fourth pluralities of conductive segments are disposed over the ground plane to form a second antenna that is configured to radiate second RF wireless power waves towards the second shelving unit such that the larger width and height of the second shelving unit is with a second coverage area of the second RF wireless power waves. A third number of conductive segments in the third plurality of conductive segments is larger than the first number of conductive segments and a fourth number of conductive segments in the fourth plurality of conductive segments is larger than the second number of conductive segments, and the second coverage area is larger than the coverage area.

In some embodiments, to create a larger coverage area more conductive segments are added. Thereby making mounting structures that have enough conductive segments to cover the entire structure of each particular shelving unit. While the example here is of two different shelving units with different mounting structures having wireless-power transmitters with different conductive segment numbers appropriate for the dimensions of those shelving units, third, fourth, fifth sixth, etc. different numbers of such mounting structures to accommodate for different shelving structures are also contemplated.

In another aspect, a backplane (e.g., a substrate with no computing components, as shown as backplane **806** in FIG. 8) includes a radio frequency (RF) wireless-power antenna (e.g., a patch antenna, optionally with a single feed) that is configured to radiate wireless power waves using a first frequency band, the RF wireless-power antenna having a perimeter that is within a perimeter of the backplane (e.g., FIG. 8 illustrates that the RF wireless-power antenna **802** having a perimeter within a perimeter of the backplane **806**). The backplane includes a first data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) configured to transmit data signals using a second frequency band distinct from the first frequency band, and the first data-communications antenna being coupled to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna (e.g., FIG. 8 illustrates first data communication antenna **810A** coupled to the backplane **806**). The backplane includes a second data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio)

configured to transmit data signals using the second frequency band, and the second data-communications antenna being coupled to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna (e.g., FIG. 8 illustrates second data communication antenna **810B** coupled to the backplane **806** perpendicular to the first data communication antenna **810A**).

In some embodiments, the wireless-power-transmitting device includes a spacer that is coupled between the RF wireless-power antenna and the backplane such that the RF wireless-power antenna is coupled to the spacer and sits above the backplane (e.g., beneath the RF wireless-power antenna and above the backplane). For example, FIG. 8 illustrates a spacer element **808**).

In some embodiments, the RF wireless-power antenna is circularly polarized such that the wireless power waves using the first frequency band are transmitted having a circular polarization (e.g., each point in a produced electromagnetic field of the wave has a constant magnitude and rotates at a constant rate in a plane perpendicular to the direction of the wave).

In some embodiments, the first data communication antenna has a first polarization (e.g., horizontal polarization) and the second communication antenna has a second polarization that is different from the first polarization (e.g., a vertical polarization). In some embodiments, the first data communication antenna with the first polarization transmits data data-communication signals to receiving devices having data communication antennas with the first polarization and the second data communication antenna with the second polarization transmits data data-communication signals to receiving devices having data communication antennas with the second polarization (e.g., in this way, receiving devices can be positioned in many different orientations yet still receive reliable data data-communication signals. For instance, wireless power receiving device is oriented such that it receives horizontally polarized data signals and wireless power receiving device is oriented such that it receives vertically polarized data signals).

In some embodiments, the first and second edges of the RF wireless-power antenna are separated by a third notched edge that separates the first and second edges and is shorter in length than the first and second edges, such that the RF wireless-power antenna has a generally quadrilateral shape (e.g., a shape where at least four of the sides are major sides (e.g., a first length), and additional other sides that are minor sides (e.g., having a second length less than the first length)) with at least one notched edge (e.g., a triangular cut out that alters the exterior shape of the quadrilateral shape) removed from a corner of the quadrilateral (e.g., a square or rectangle antenna would have the upper-left corner notched (e.g., a triangular notch) and the bottom-right corner notched). For example, FIG. 8 illustrates two notched edges (e.g., a first notch **804A** and a second notch **804B**).

In some embodiments, the RF wireless-power antenna has two notched edges, the two notched edges being symmetrically shaped (e.g., FIG. 8 illustrates that the first notch **804A** and the second notch **804B** are symmetrical to each other).

In some embodiments, the first communication antenna is placed in a middle of the first edge of the perimeter of the RF wireless-power antenna, and the second communication antenna is placed in a middle of the second edge of the perimeter of the RF wireless-power antenna. FIG. 8 illustrates that first data communication antenna **810A** is placed

in the middle of the first edge **812A** of the perimeter of the RF wireless-power antenna, and the second data communication antenna **810A** is placed in the middle of the second edge **812A** of the perimeter of the RF wireless-power antenna.

In some embodiments, wherein the RF wireless-power antenna is a patch antenna. In some embodiments, the RF wireless-power antenna is a stamped metal patch antenna.

In some embodiments, the patch antenna is constructed of copper material. In some embodiments, the patch antenna is an alloy that includes copper, steel, aluminum.

In some embodiments, a peak gain of the RF wireless-power antenna is greater than 8 dBi while the first frequency band is a center frequency band of 918 MHz.

In some embodiments, the wireless-power antenna matching is less than -10 dB.

In some embodiments, the first data-communications antenna and the second data-communications antenna produce a peak gain of 5 dBi while operating at the second frequency band of approximately 2.45 GHz. In some embodiments, the first data-communications antenna and second data-communications antenna independently produce a peak gain of 5 dBi at 2.45 GHz.

In some embodiments, the first data-communications antenna and the second data-communications antenna have a matching of greater than -7 dB.

In some embodiments, the RF frequency wireless power antenna improves the gain of both the first data-communications antenna and the second data-communications antenna as a result of their proximity to the wireless-power antenna (e.g., a gain of 1 dB before being near wireless-power antenna and gain of 4 dB after being near wireless-power antenna).

In some embodiments, the first data-communications antenna and the second data-communications antenna use the RF frequency wireless power antenna and a main ground to provide maximum gain and optimum radiation performance of the first data-communications antenna and the second data-communications antenna.

In some embodiments, the first data-communications antenna and the second data-communications antenna are formed using respective stamped copper plates.

In some embodiments, the first data-communications antenna and the second data-communications antenna are each suspended 0.1 to 0.5 inches from a top surface of the backplane (e.g., FIG. 8 illustrates that the first data communication antenna **810A** is suspended by post **814A** and the second data communication antenna **810B** is suspended by post **814B**). In some embodiments, the first data-communications antenna and the second data-communications antenna are suspended by a metallic pin/post coupled to the backplane.

In another aspect, a wireless-power transmitting and receiving system, includes a plurality of wireless power-transmitting devices including any of the shelving-mounted wireless power transmitting devices discussed above, and including a plurality of wireless-power harvesters structured in accordance with any of discussion above.

In another aspect, a method of manufacturing a wireless-power device comprises providing a backplane is performed (e.g., a substrate with no computing components) (e.g., a substrate with no computing components, as shown as backplane **806** in FIG. 8). The method of manufacturing the wireless-power device comprises disposing, on the backplane, a radio frequency (RF) wireless-power antenna that is configured to radiate wireless power waves using a first frequency band, the RF wireless-power antenna having a

perimeter that is within a perimeter of the backplane (e.g., FIG. 8 illustrates that the RF wireless-power antenna **802** having a perimeter within a perimeter of the backplane **806**). The method of manufacturing the wireless-power device comprises coupling a first data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna, wherein the first data-communications antenna transmits data signals using a second frequency band distinct from the first frequency band (e.g., FIG. 8 illustrates first data communication antenna **810A** coupled to the backplane **806**). The method of manufacturing the wireless-power device comprises coupling a second data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna, and further wherein the second data-communications antenna transmit data signals using the second frequency band (e.g., FIG. 8 illustrates second data communication antenna **810B** coupled to the backplane **806** perpendicular to the first data communication antenna **810A**).

The preceding description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the embodiments described herein and variations thereof. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the subject matter disclosed herein. Thus, the present disclosure is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the following claims and the principles and novel features disclosed herein.

Features of the present invention can be implemented in, using, or with the assistance of a computer program product, such as a storage medium (media) or computer readable storage medium (media) having instructions stored thereon/ in which can be used to program a processing system to perform any of the features presented herein. The storage medium (e.g., memory **1006** and **1034** in FIG. 10) can include, but is not limited to, high-speed random access memory, such as DRAM, SRAM, DDR RAM or other random access solid state memory devices, and may include non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory optionally includes one or more storage devices remotely located from the CPU(s) (e.g., processor(s)). Memory, or alternatively the non-volatile memory device(s) within the memory, comprises a non-transitory computer readable storage medium.

Stored on any one of the machine readable medium (media), features of the present invention can be incorporated in software and/or firmware for controlling the hardware of a processing system (such as the components associated with the wireless-power transmitter **1035** and/or wireless-power receivers **1055**), and for enabling a processing system to interact with other mechanisms utilizing the results of the present invention. Such software or firmware may include, but is not limited to, application code, device drivers, operating systems, and execution environments/containers.

It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various

elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the claims. As used in the description of the embodiments and the appended claims, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

As used herein, the term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in accordance with a determination” or “in response to detecting,” that a stated condition precedent is true, depending on the context. Similarly, the phrase “if it is determined [that a stated condition precedent is true]” or “if [a stated condition precedent is true]” or “when [a stated condition precedent is true]” may be construed to mean “upon determining” or “in response to determining” or “in accordance with a determination” or “upon detecting” or “in response to detecting” that the stated condition precedent is true, depending on the context.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the claims to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain principles of operation and practical applications, to thereby enable others skilled in the art.

What is claimed is:

1. A wireless-power transmitting system, comprising:
 - a wireless-power transmitter coupled within an enclosed environment, wherein: the enclosed environment includes one or more shelving units, and
 - the wireless-power transmitter is configured to radiate radio-frequency (RF) wireless-power waves towards the one or more shelving units using a first frequency band, the wireless-power transmitter includes:
 - a backplane with one or more data-communications antennas configured to transmit data signals using a second frequency band distinct from the first frequency band,
 - a ground plane,
 - a first plurality of conductive segments having a first shape and a first size,
 - a second plurality of conductive segments having a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size, wherein a respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another, and
 - the first and second pluralities of conductive segments being disposed over the ground plane to form an antenna that is configured to radiate RF wireless-power waves; and

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a wireless-power receiver integrated in a small form-factor device coupled to or on a portion of the shelving unit, wherein:

the wireless-power receiver comprises an antenna and power-conversion circuitry that is configured to convert the RF wireless-power waves harvested by the antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device.

2. The wireless-power transmitting system of claim 1, wherein the wireless-power transmitter is coupled to a supporting structure of the enclosed environment and configured to radiate RF wireless-power waves towards a plurality of the one or more shelving units in the enclosed environment.

3. The wireless-power transmitting system of claim 1, wherein the small form-factor device is configured to at least one of transmit, receive, or display data.

4. The wireless-power transmitting system of claim 1, wherein the enclosed environment includes at least one of: a room, a building, a warehouse, a cargo compartment, or a delivery compartment.

5. The wireless-power transmitting system of claim 1, wherein the antenna of the wireless-power receiver is a stamped metal harvesting antenna coupled to a PCB that includes components for operating and powering the small form-factor device.

6. The wireless-power transmitting system of claim 5, wherein the stamped metal harvesting antenna is quasi omnidirectional and configured to receive RF wireless power waves having any polarization.

7. The wireless-power transmitting system of claim 1, wherein one or more shelving units include one or more shelves and a respective wireless-power receiver is coupled to each one of the one or more shelves.

8. The wireless-power transmitting system of claim 1, wherein the wireless-power transmitter comprises a radio frequency (RF) wireless-power antenna that is circularly polarized such that the radiated RF power waves are transmitted having a circular polarization.

9. The wireless-power transmitting system of claim 1, wherein the first shape and the second shape are a meandering path that produces a rectangular shape.

10. A method, comprising:
radiating, via a wireless-power transmitter coupled within an enclosed environment, RF wireless-power waves

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towards one or more shelving units within the enclosed environment, the wireless-power transmitter including:
a ground plane,

a first plurality of conductive segments having a first shape and a first size,

a second plurality of conductive segments having a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size, wherein a respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another, and

the first and second pluralities of conductive segments being disposed over the ground plane to form an antenna that is configured to radiate RF wireless-power waves;

wherein:

the RF wireless-power waves are configured to be received by a wireless-power receiver coupled to or on a portion of a shelving unit of one or more shelving units, and

the wireless-power receiver that is configured to convert the RF wireless-power waves, harvested by an antenna of the wireless-power receiver, into usable energy for charging a battery of a small form-factor device or for powering the small form-factor device.

11. The method of claim 10, wherein the wireless-power transmitter is coupled to a supporting structure of the enclosed environment and configured to radiate RF wireless-power waves towards a plurality of the one or more shelving units in the enclosed environment.

12. The method of claim 10, wherein the small form-factor device is configured to at least one of transmit, receive, or display data.

13. The method of claim 10, wherein the enclosed environment includes at least one of:

a room, a building, a warehouse, a cargo compartment, or a delivery compartment.

14. The method of claim 10, wherein:

the RF wireless-power waves are configured to be radiated using a first frequency band; and

the wireless-power transmitter comprises a backplane including one or more data-communications antennas configured to transmit data signals using a second frequency band distinct from the first frequency band.

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